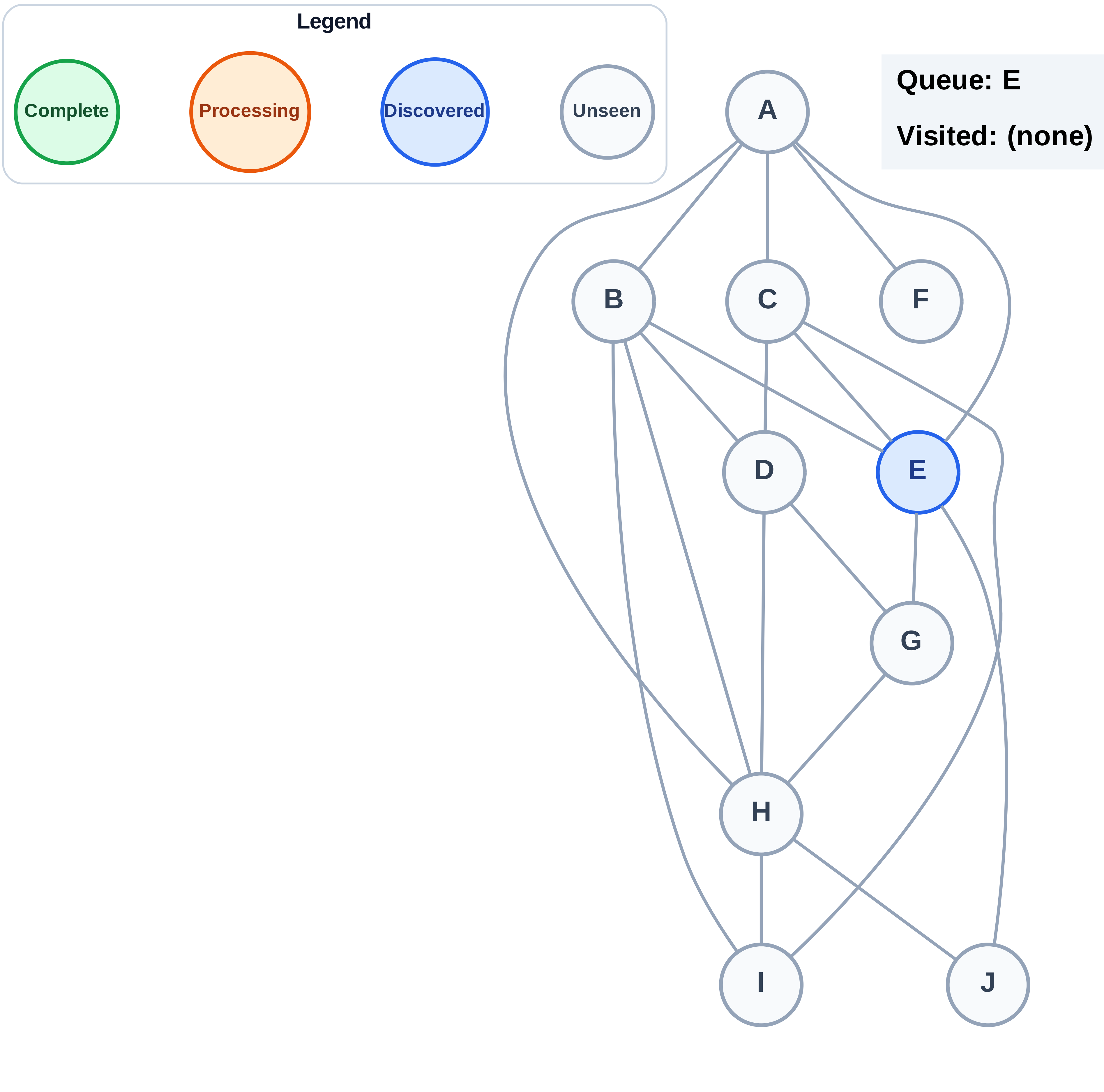
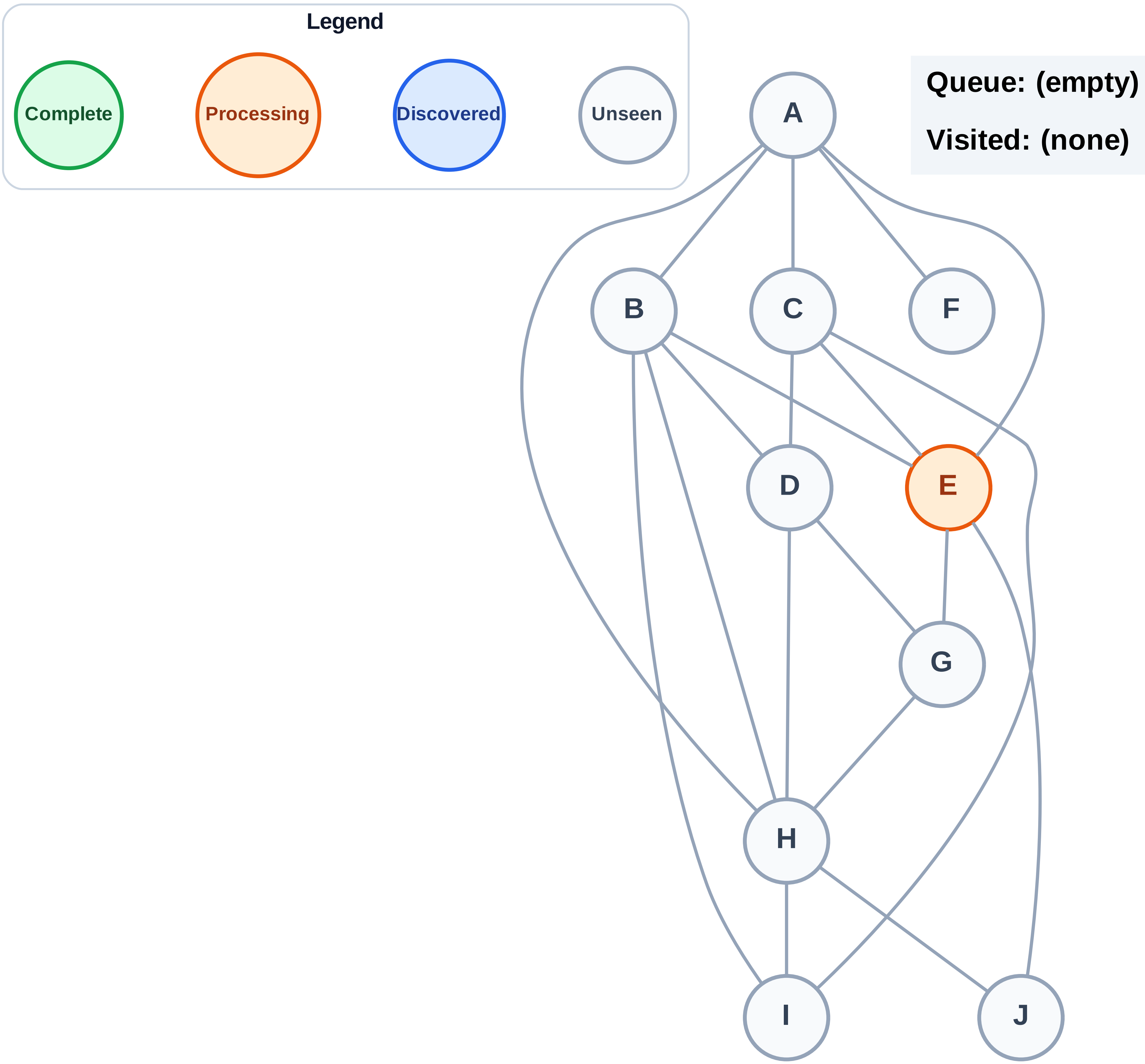
BFS Traversal

Step 1: Start at E



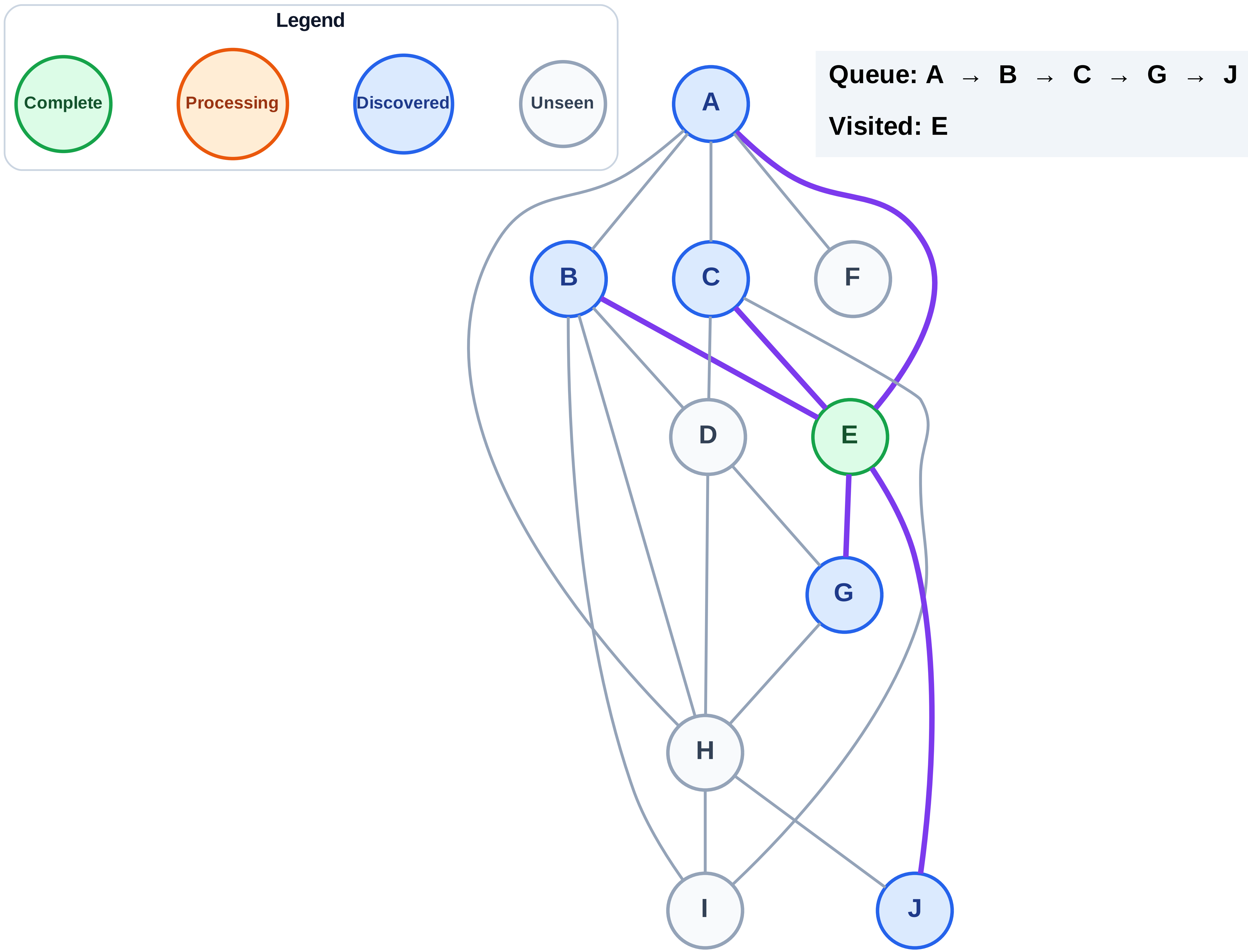
BFS Traversal

Step 2: Process node E



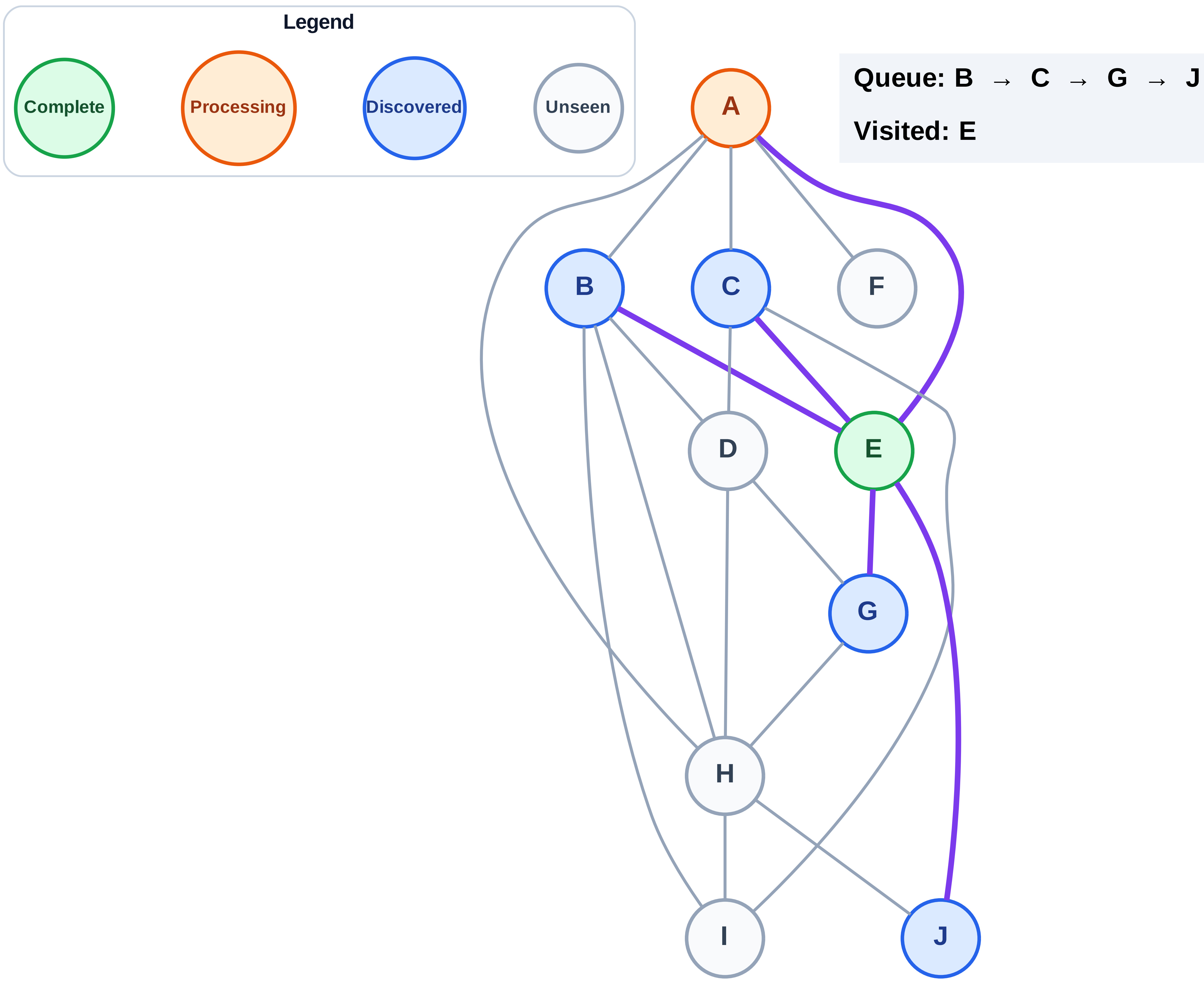
BFS Traversal

Step 3: Discover A, B, C, G, J from E



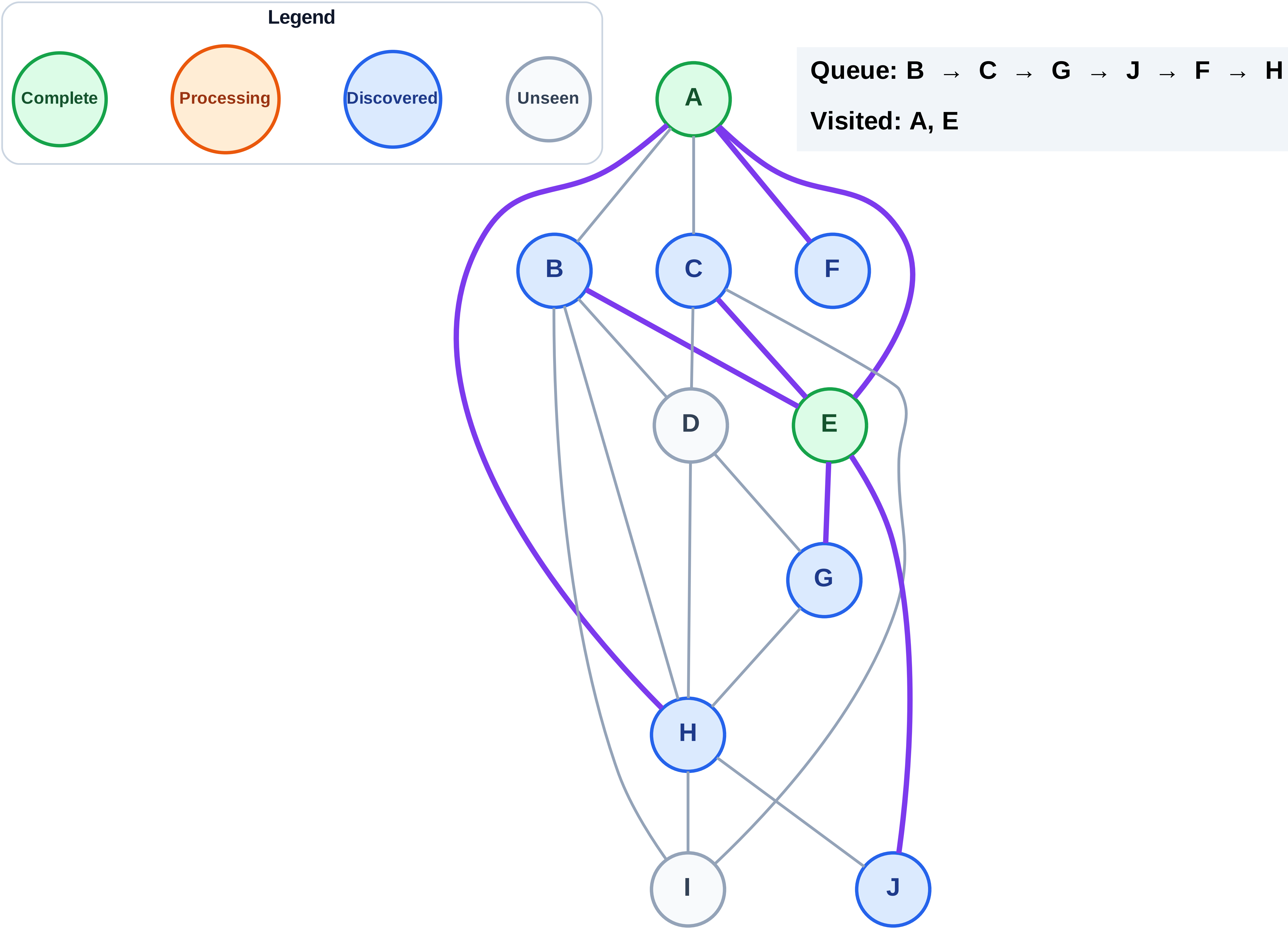
BFS Traversal

Step 4: Process node A



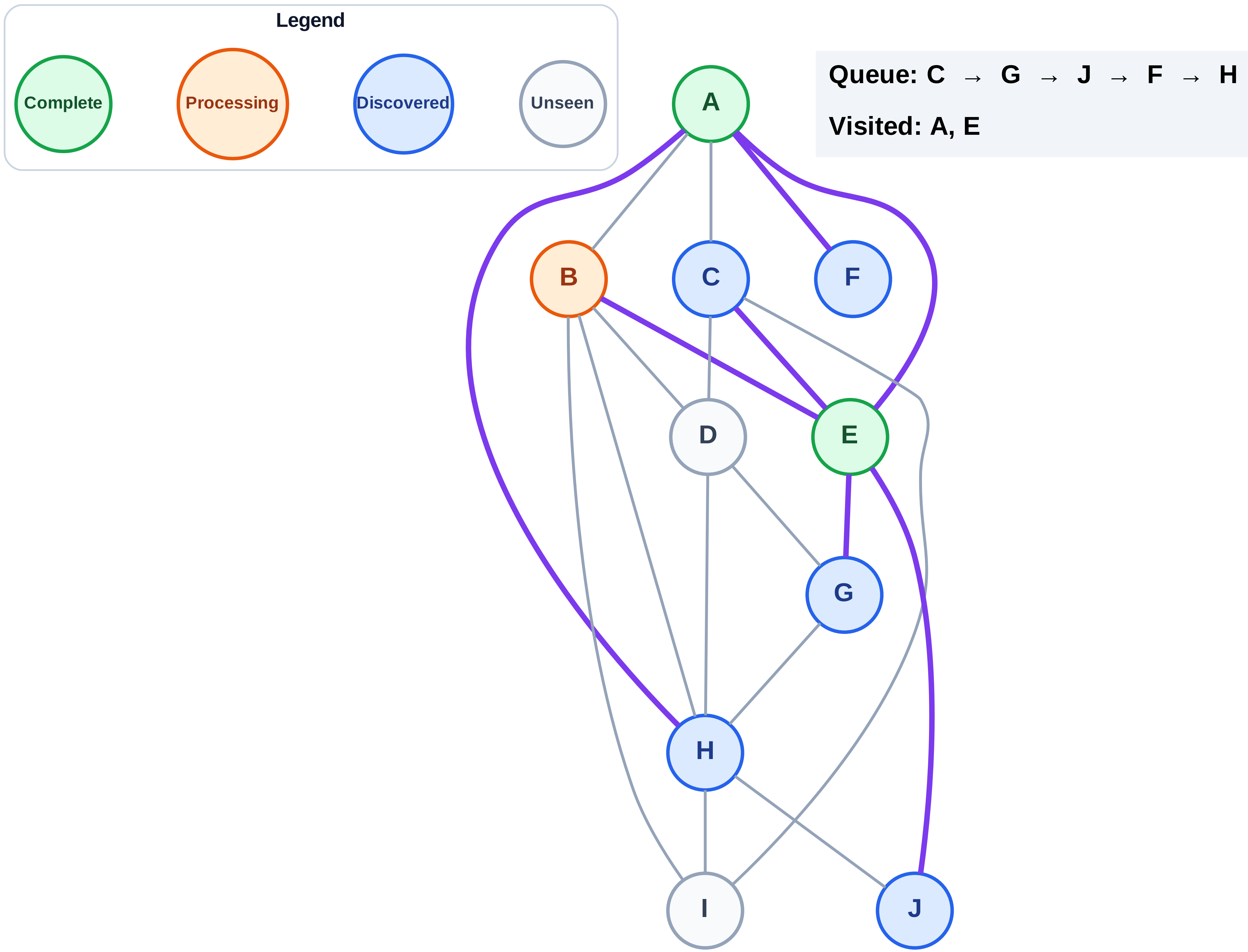
BFS Traversal

Step 5: Discover F, H from A



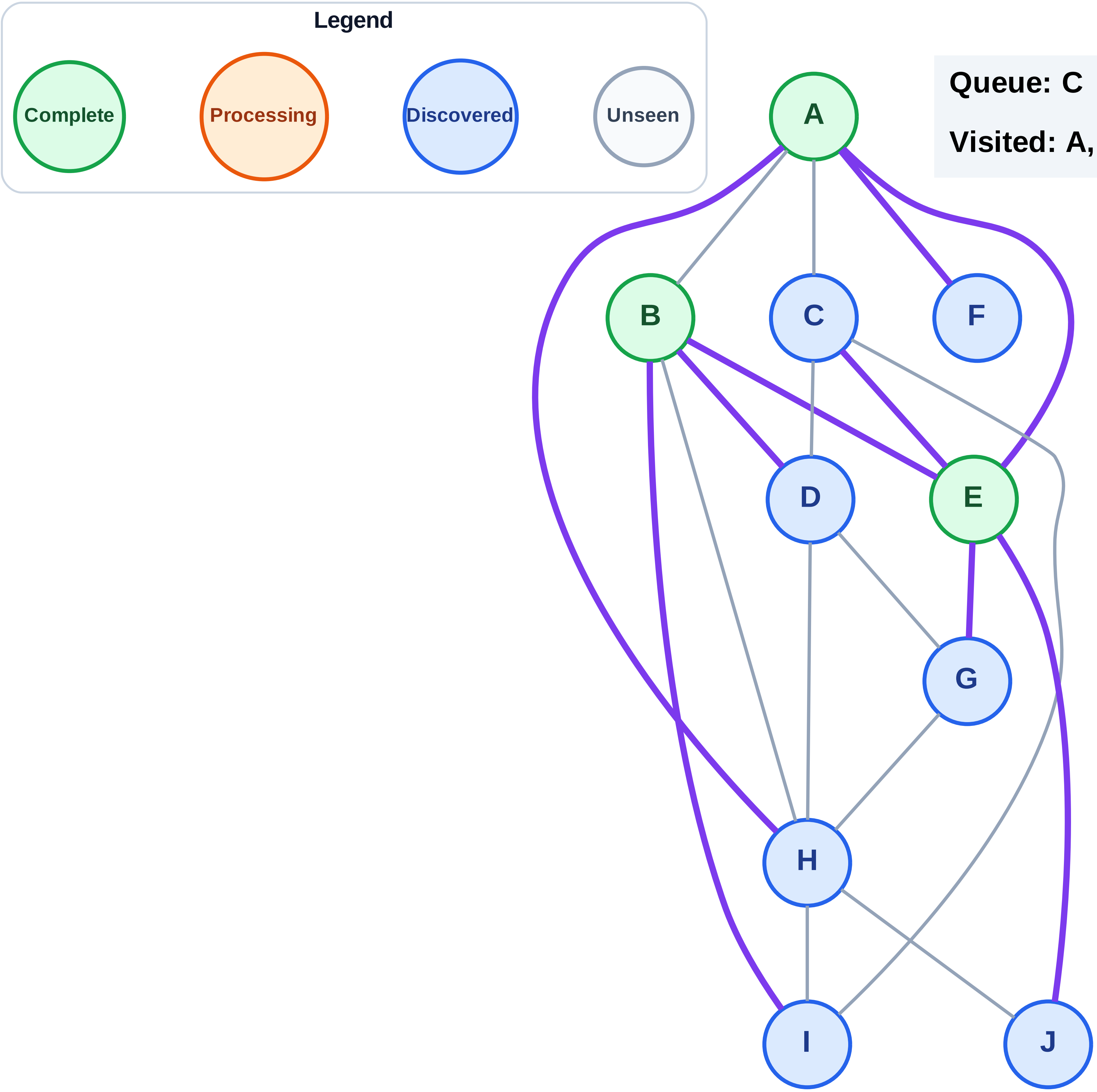
BFS Traversal

Step 6: Process node B



BFS Traversal

Step 7: Discover D, I from B

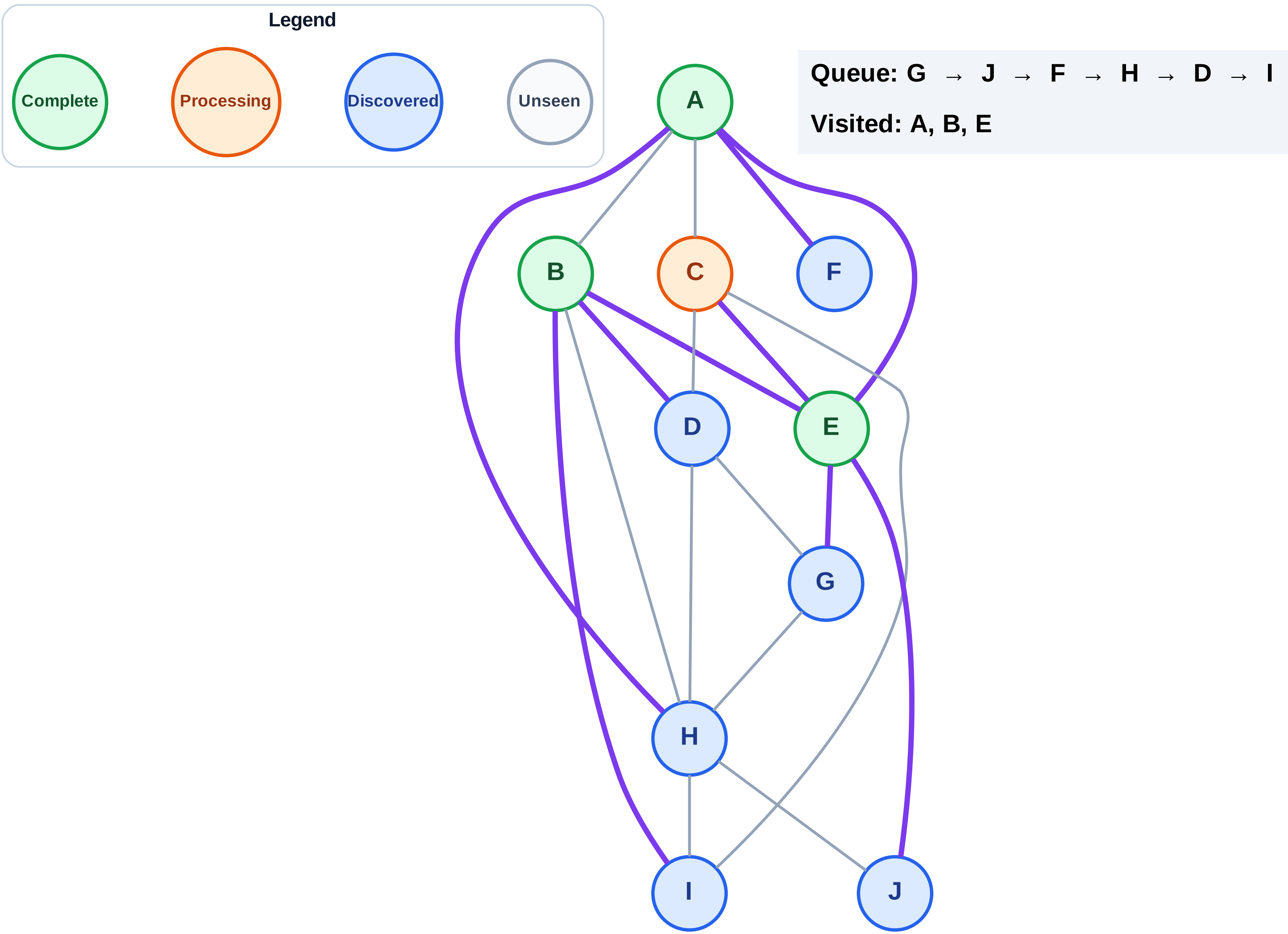


Queue: C → G → J → F → H → D → I

Visited: A, B, E

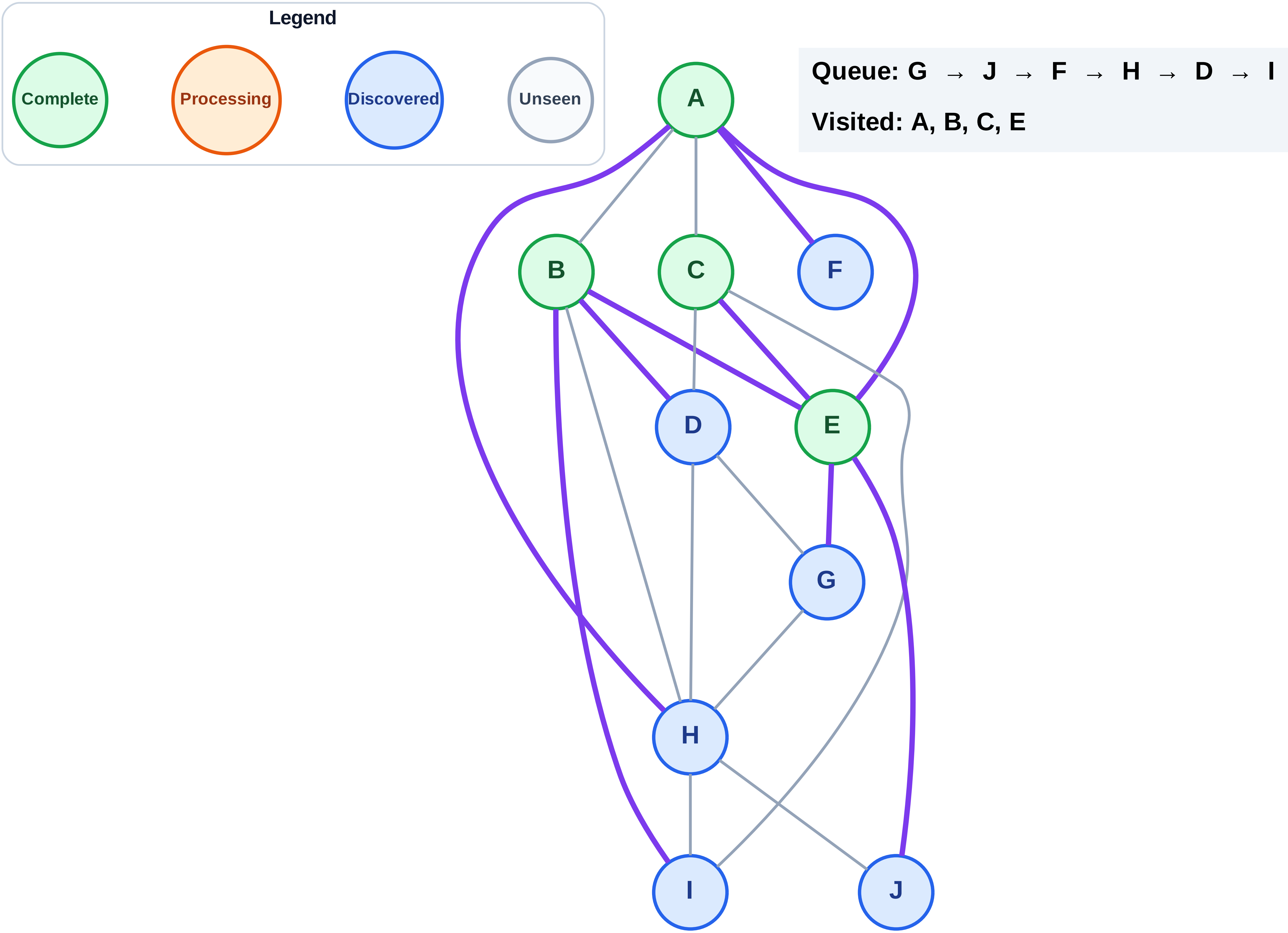
BFS Traversal

Step 8: Process node C



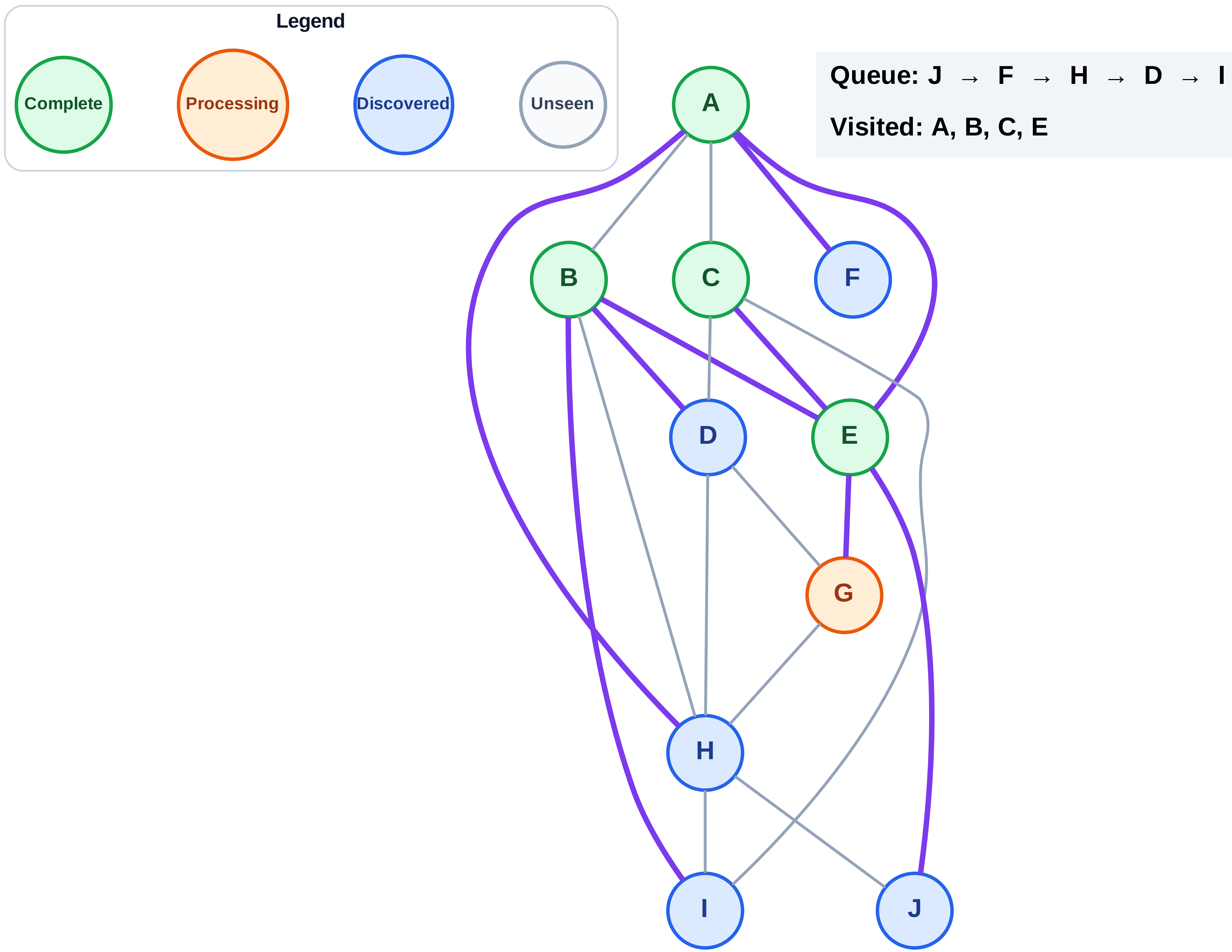
BFS Traversal

Step 9: No new neighbors from C



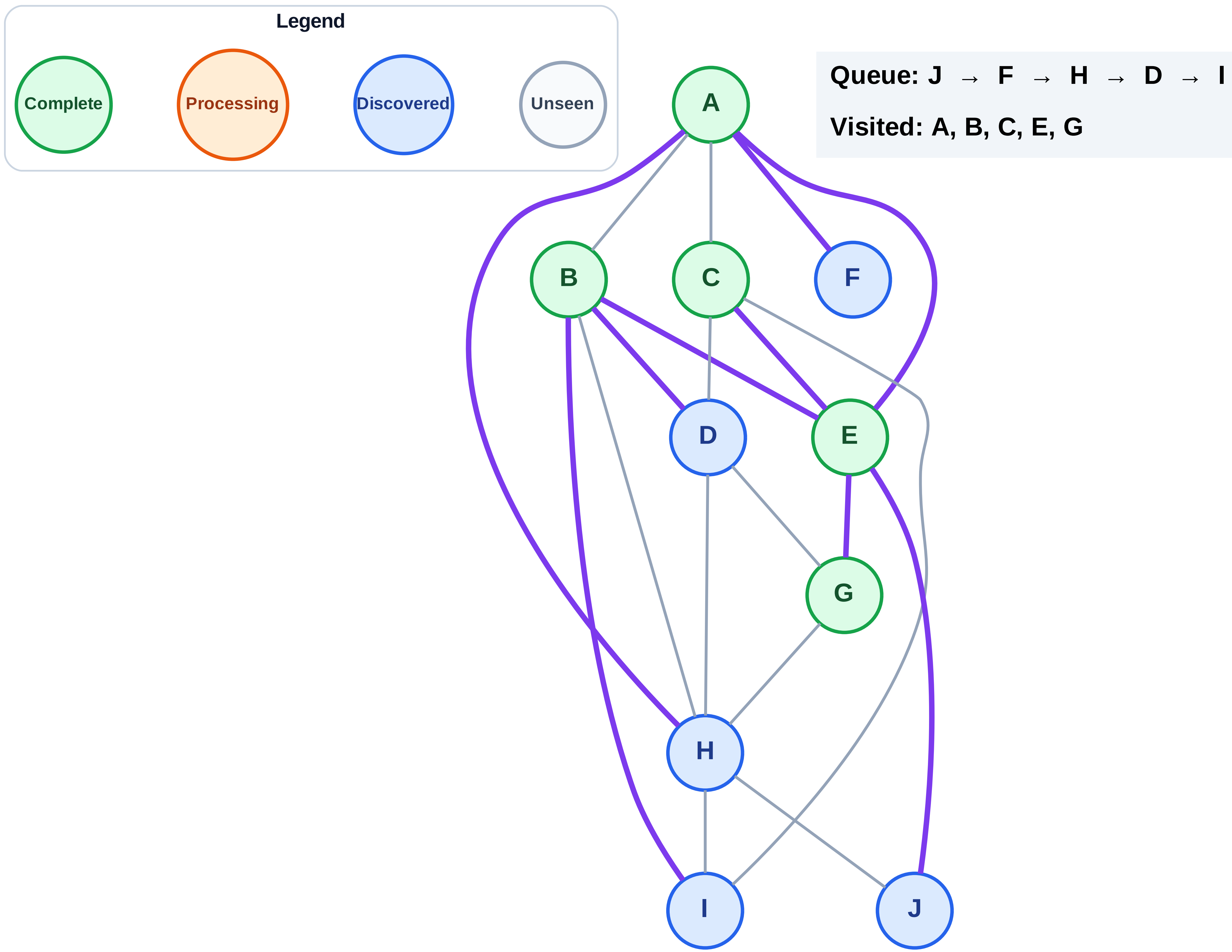
BFS Traversal

Step 10: Process node G



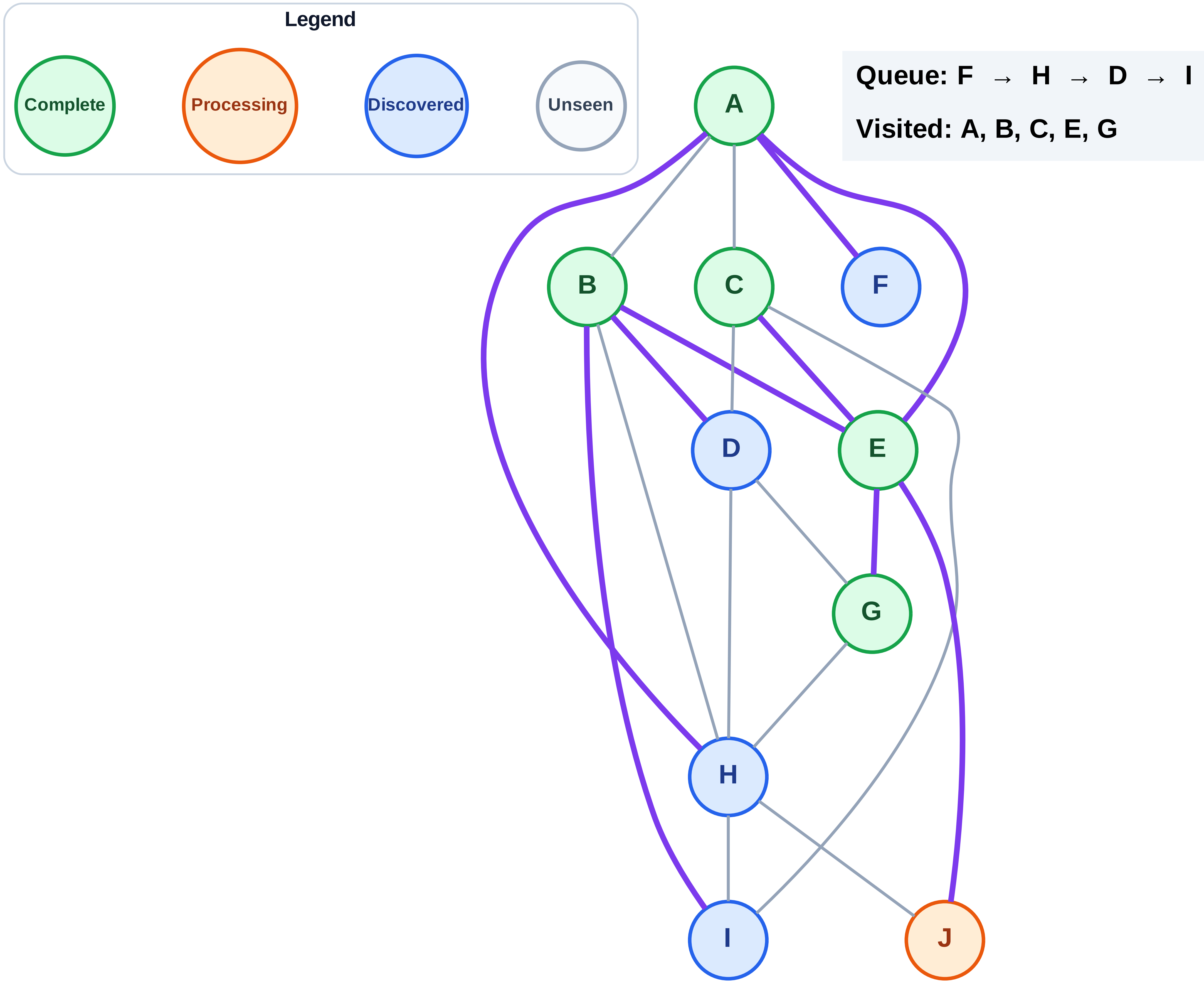
BFS Traversal

Step 11: No new neighbors from G



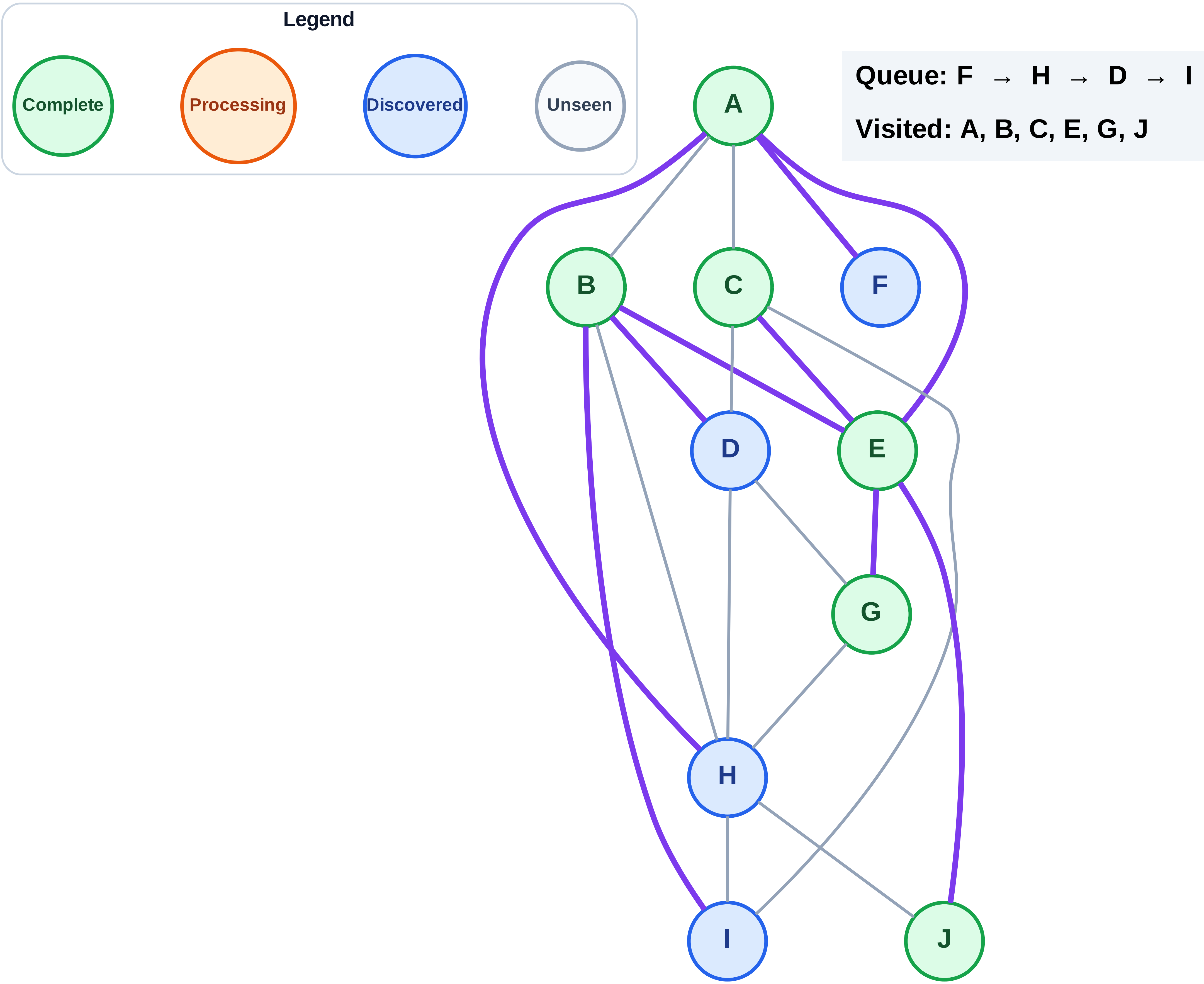
BFS Traversal

Step 12: Process node J



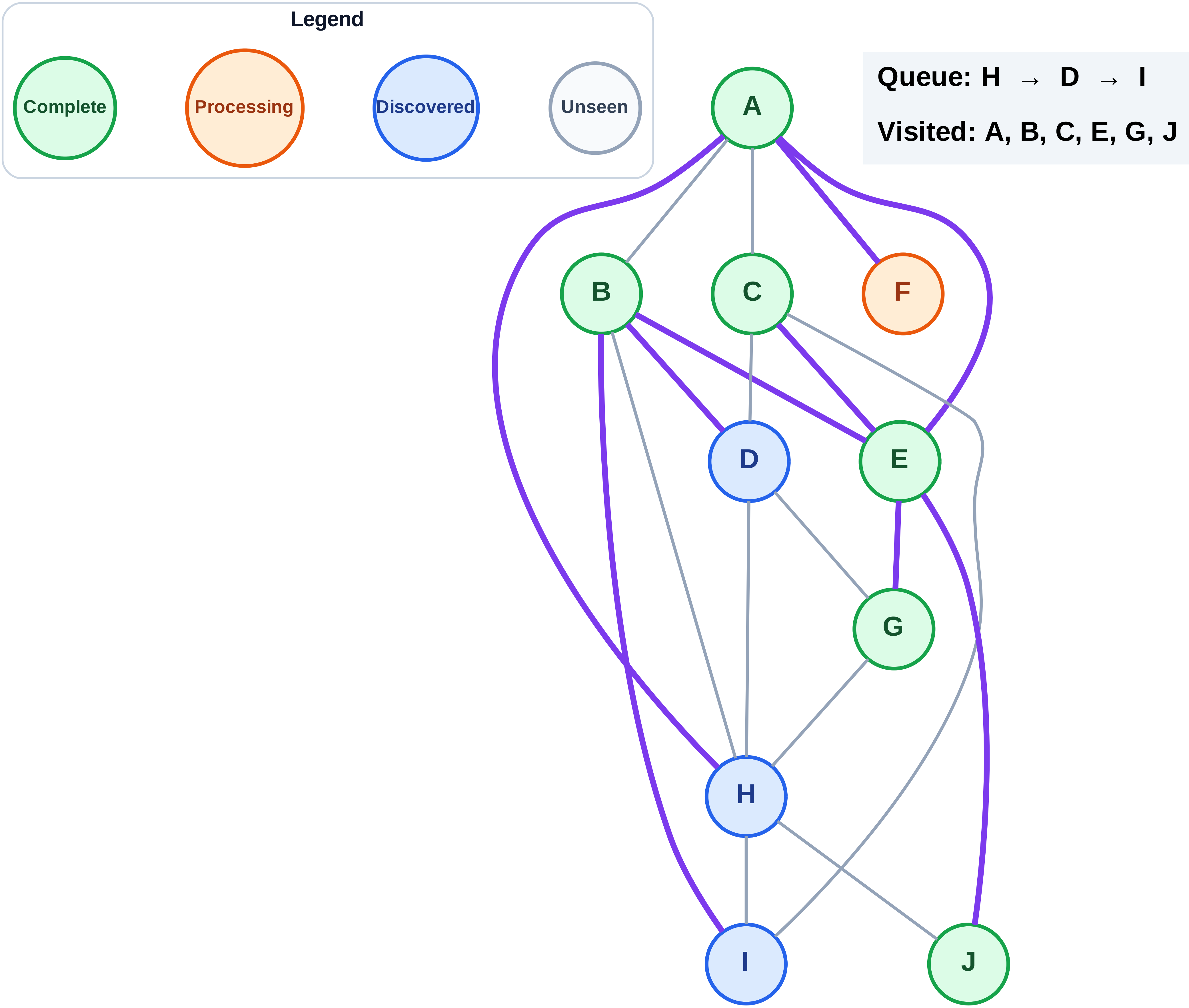
BFS Traversal

Step 13: No new neighbors from J



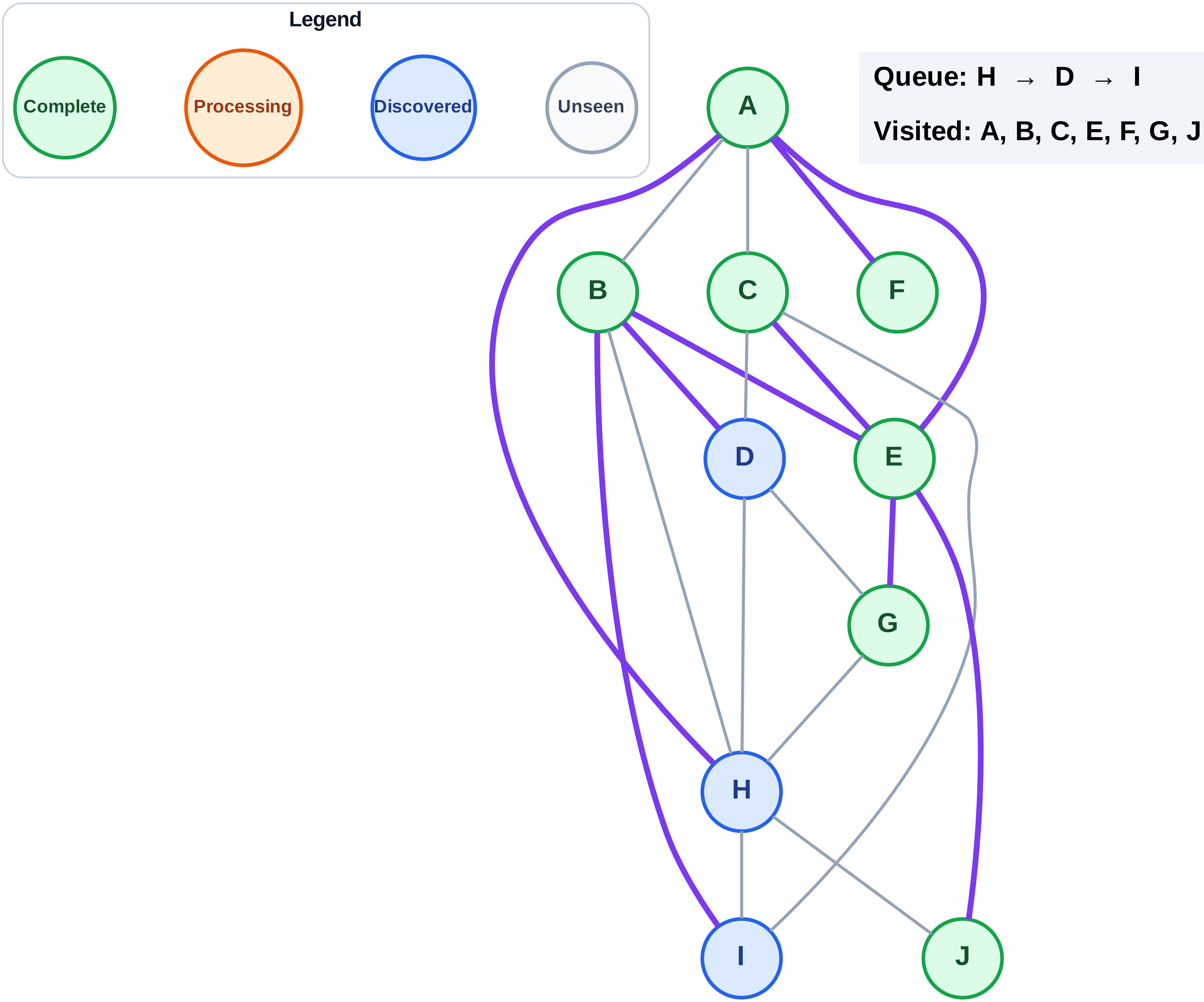
BFS Traversal

Step 14: Process node F



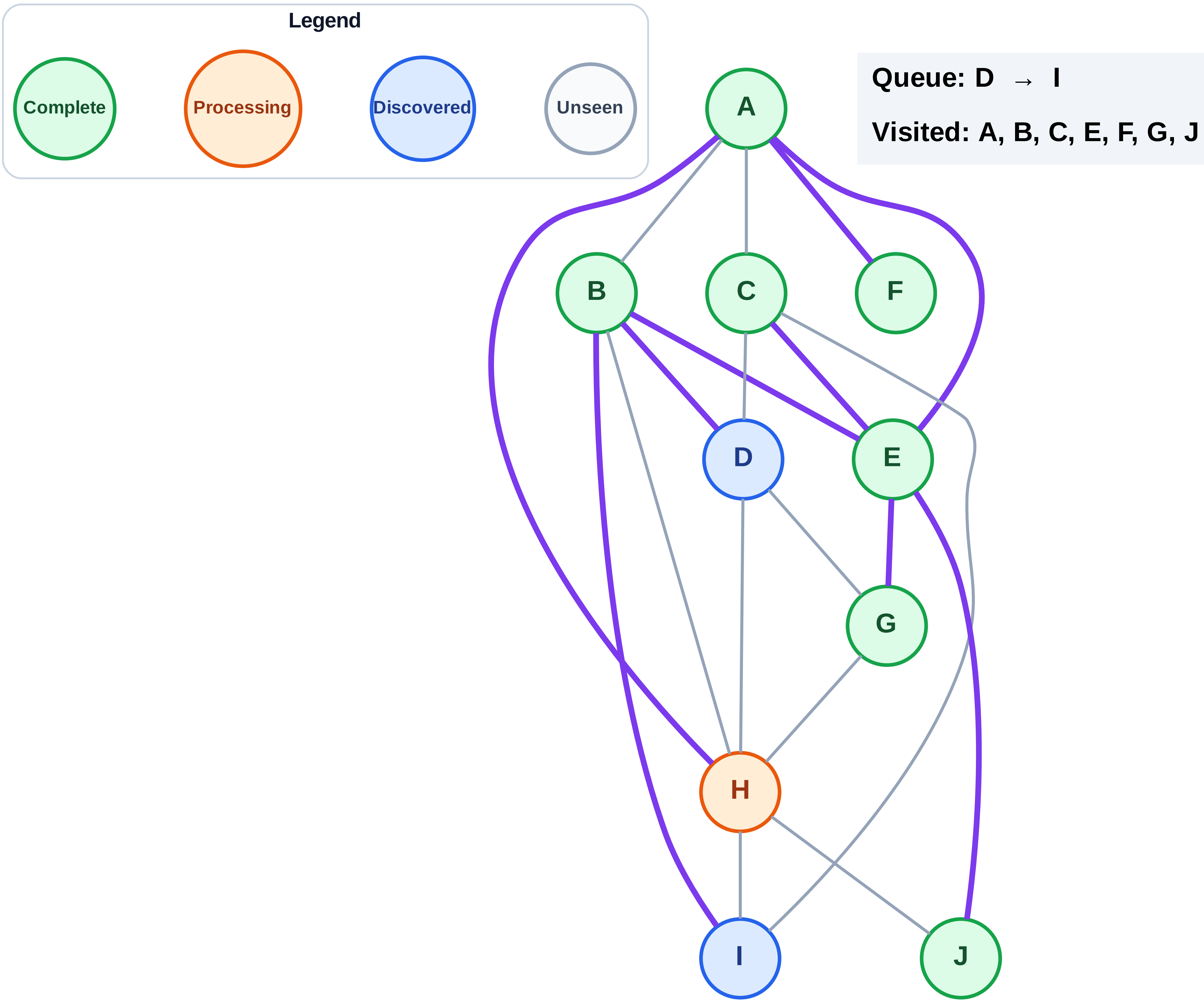
BFS Traversal

Step 15: No new neighbors from F



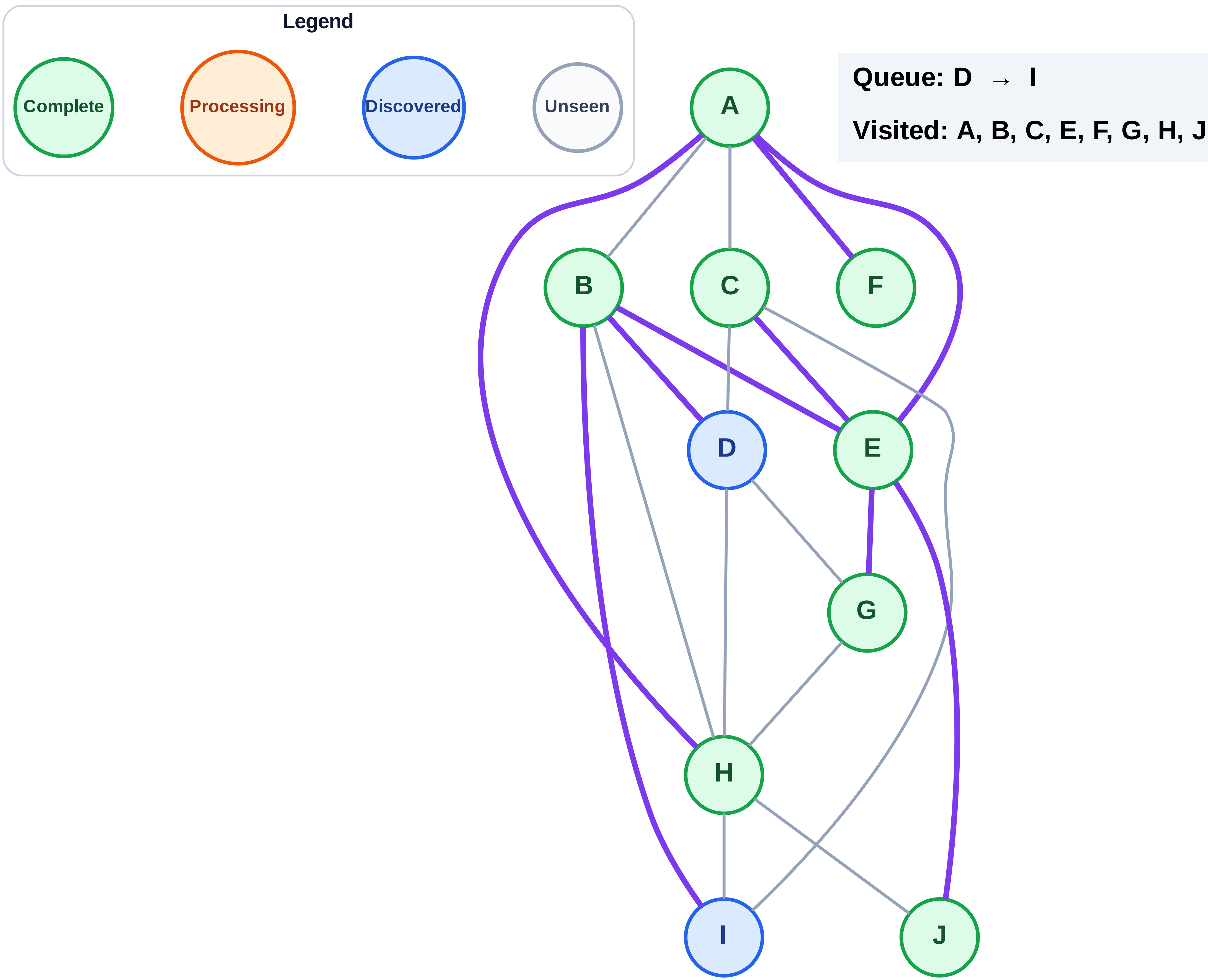
BFS Traversal

Step 16: Process node H



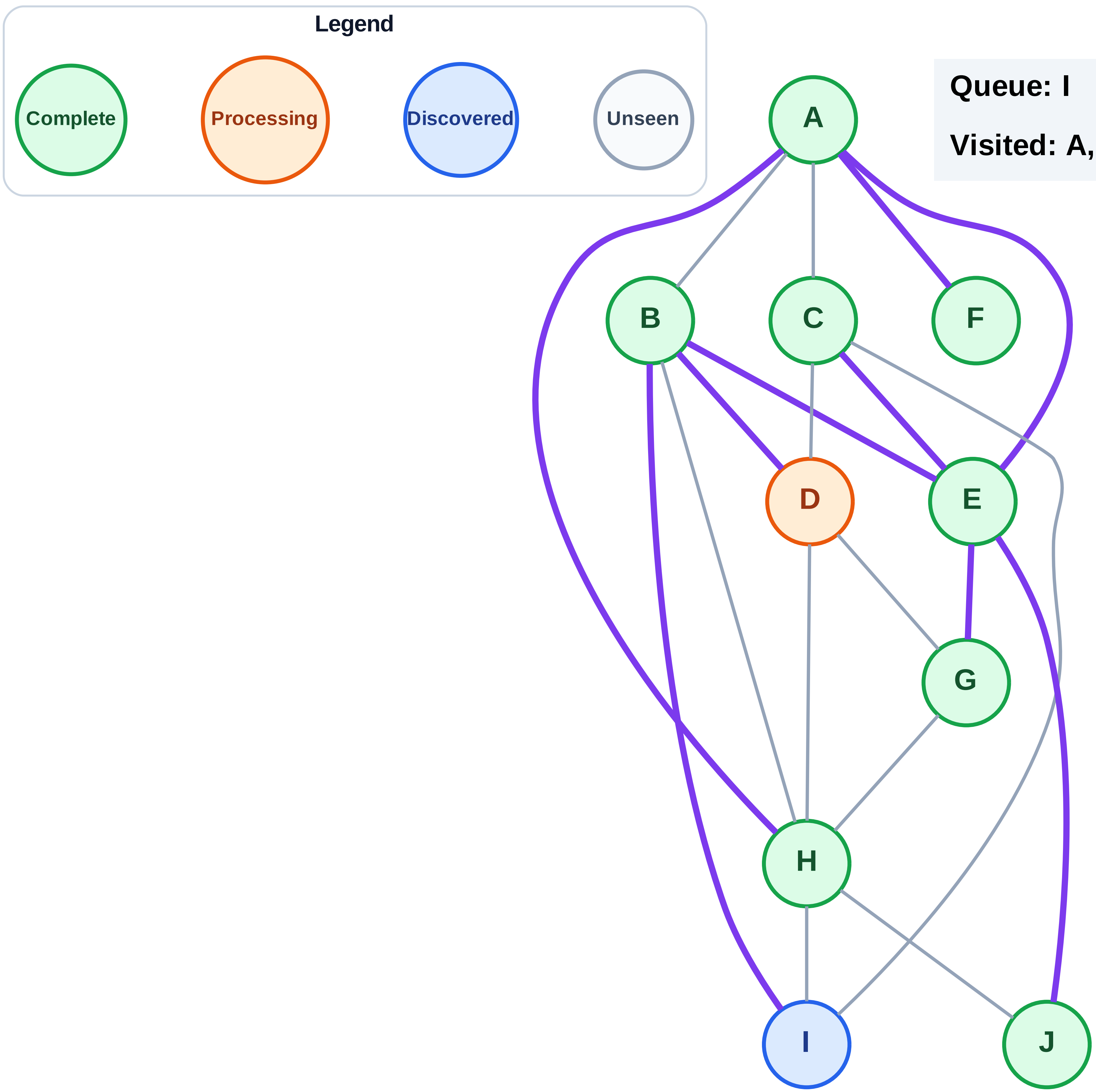
BFS Traversal

Step 17: No new neighbors from H



BFS Traversal

Step 18: Process node D

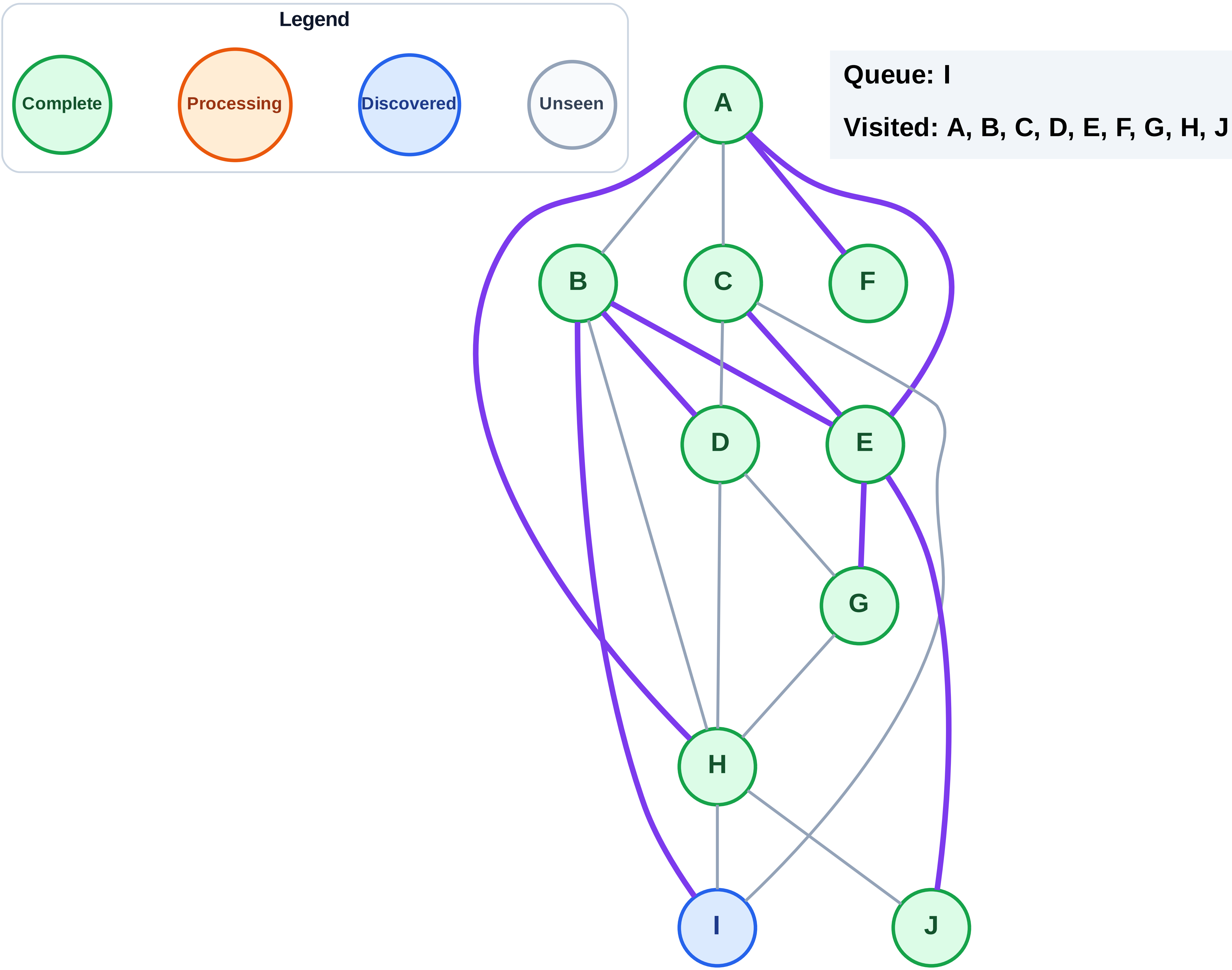


Queue: I

Visited: A, B, C, E, F, G, H, J

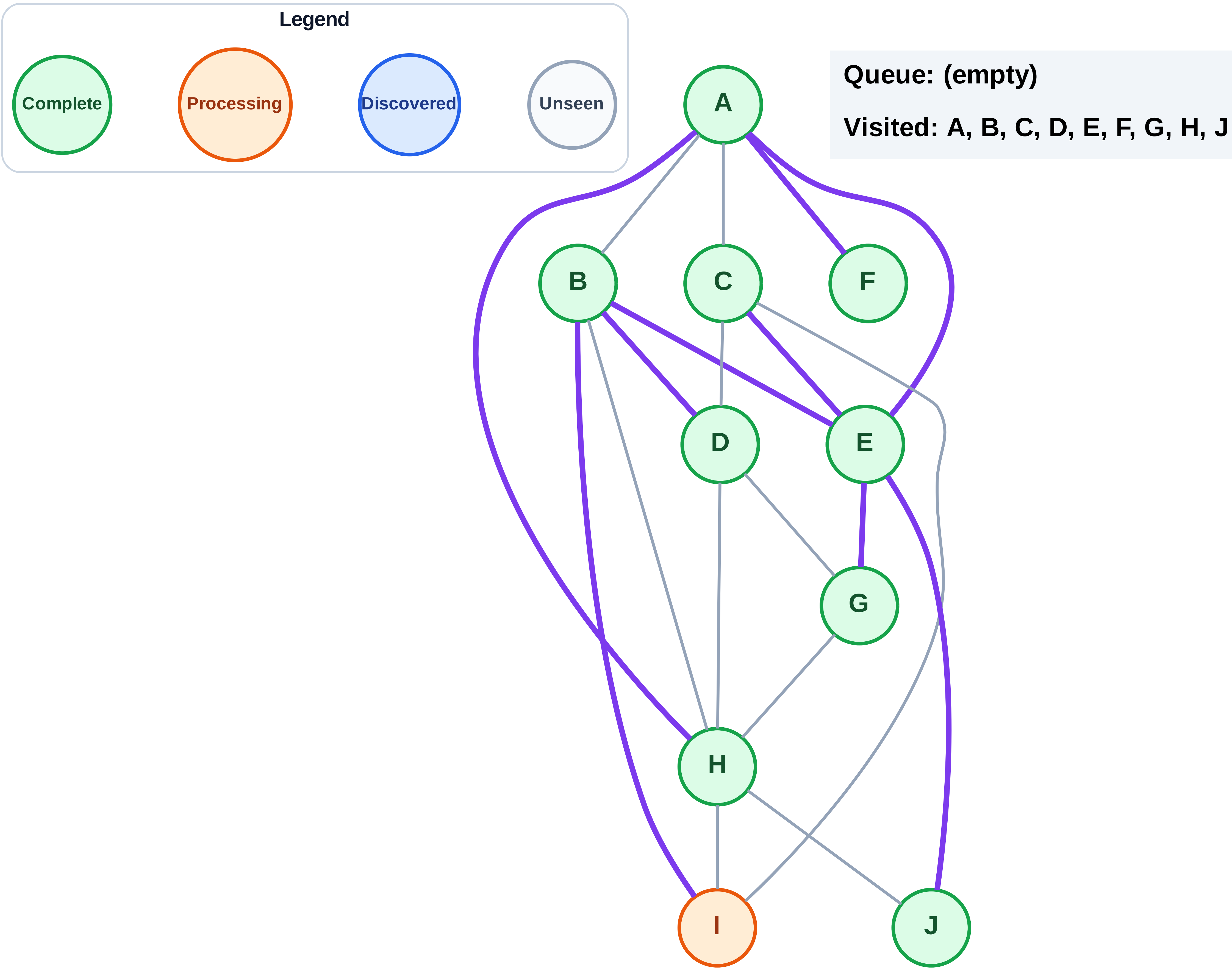
BFS Traversal

Step 19: No new neighbors from D



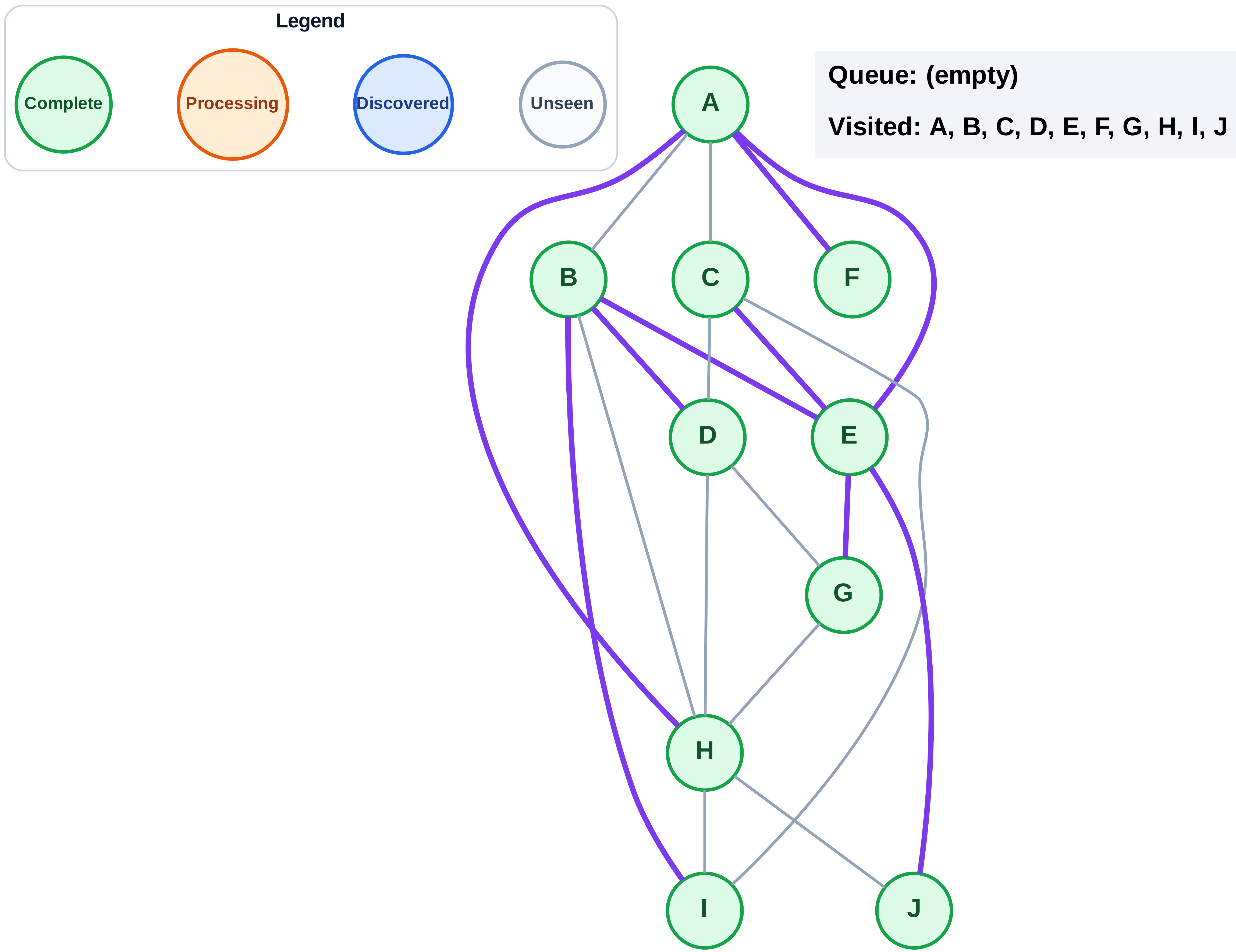
BFS Traversal

Step 20: Process node I



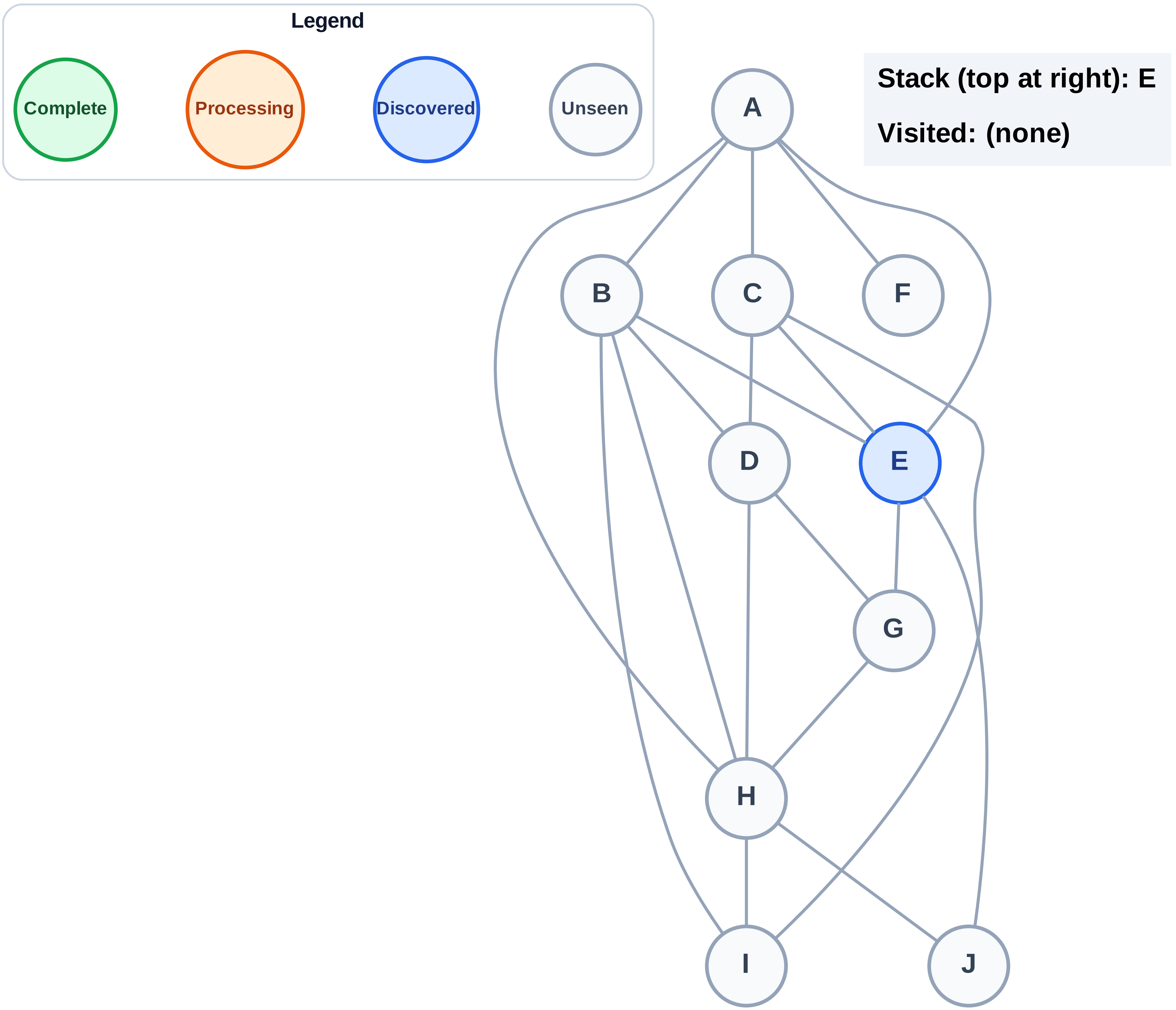
BFS Traversal

Step 21: No new neighbors from I



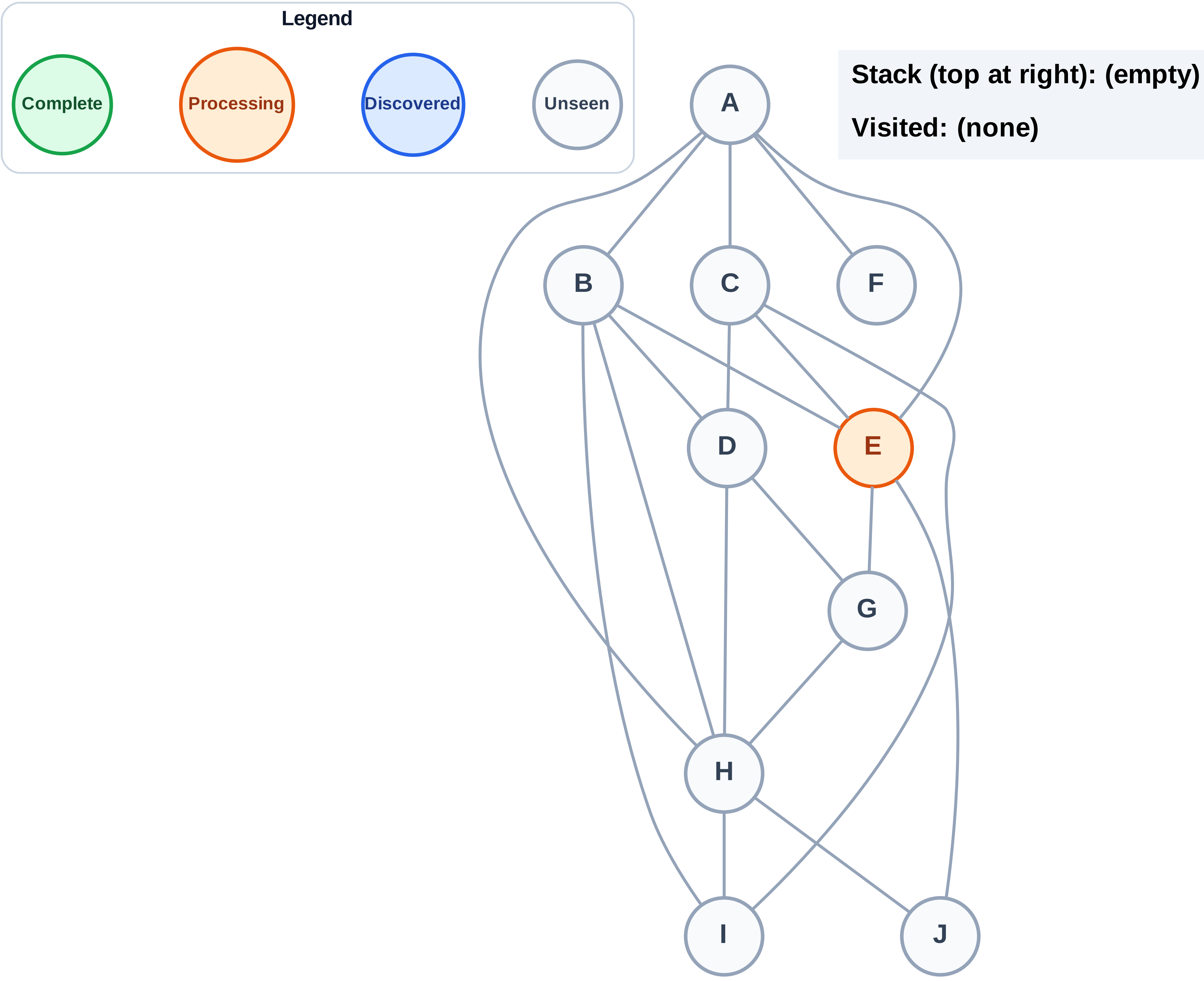
DFS Traversal

Step 1: Start at E



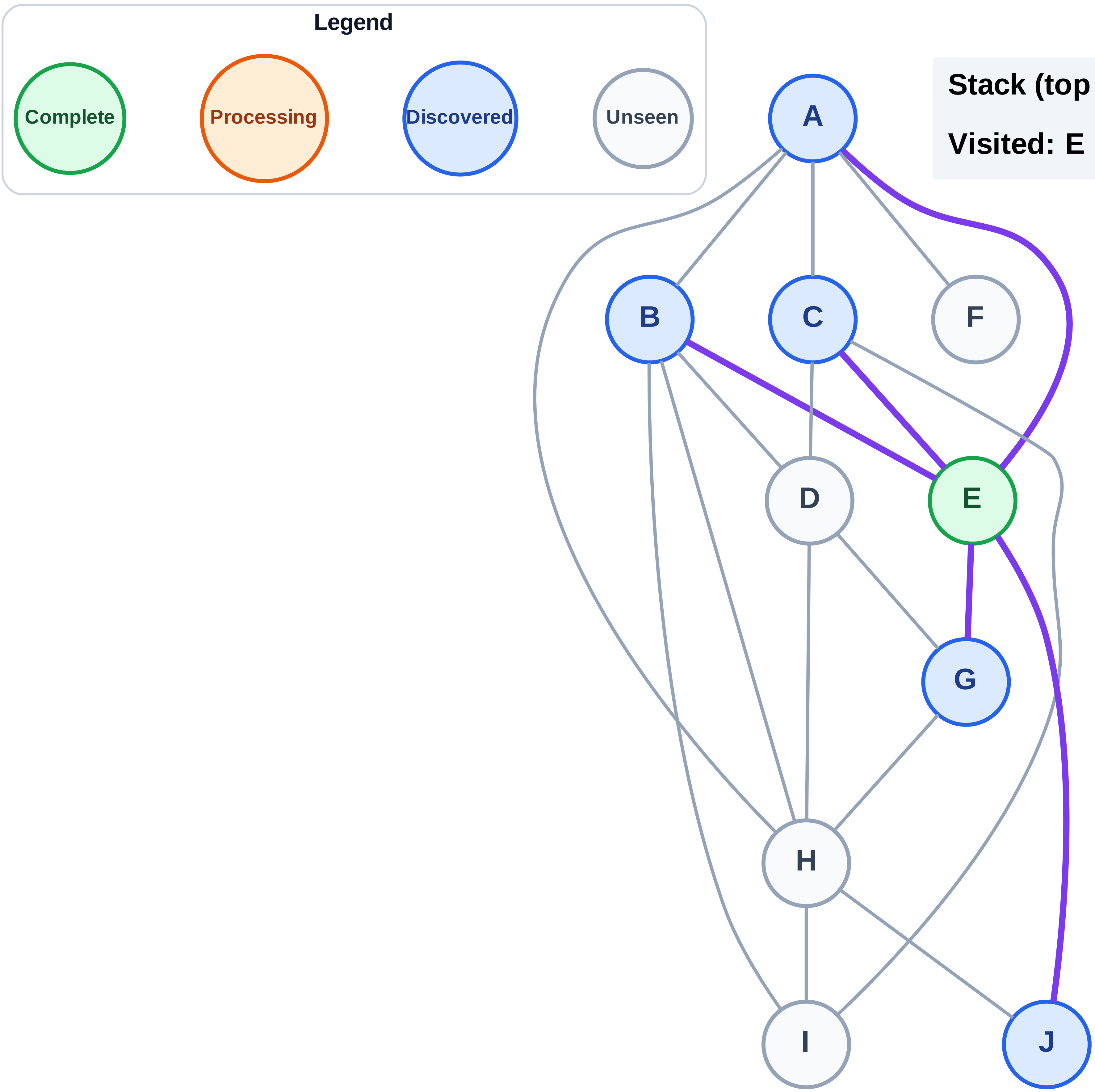
DFS Traversal

Step 2: Process node E



DFS Traversal

Step 3: Discover J, G, C, B, A from E

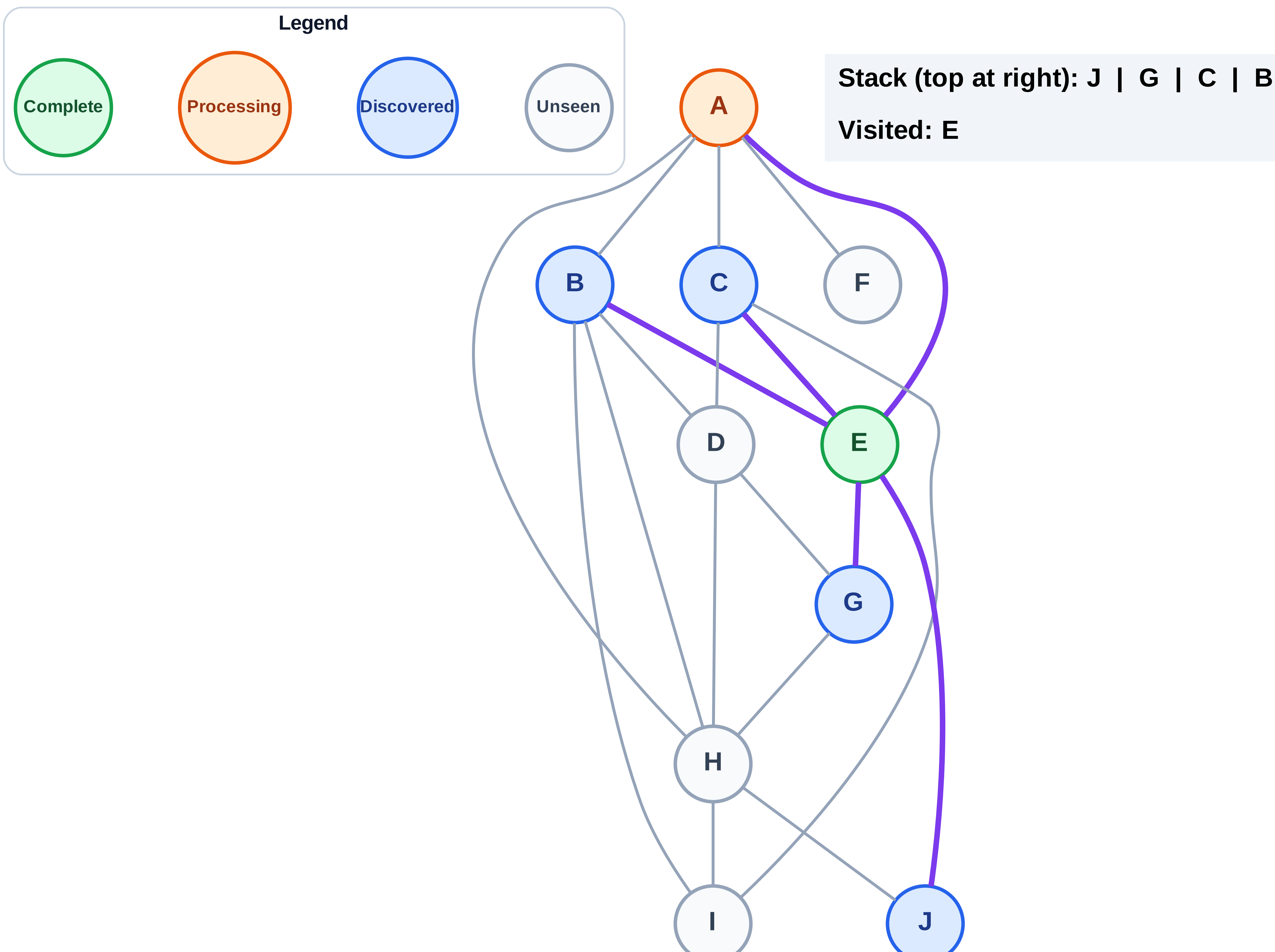


Stack (top at right): J | G | C | B | A

Visited: E

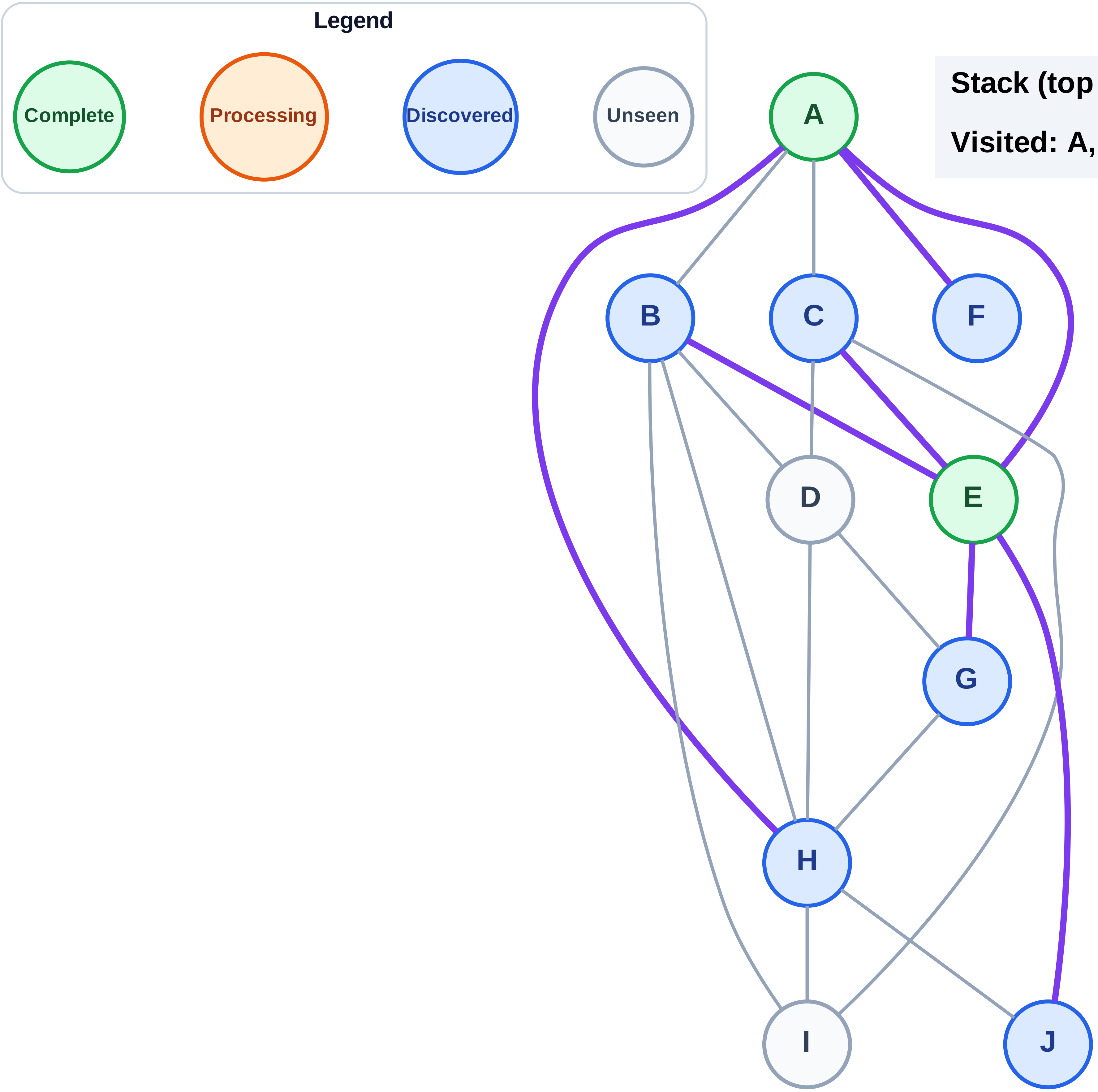
DFS Traversal

Step 4: Process node A



DFS Traversal

Step 5: Discover H, F from A

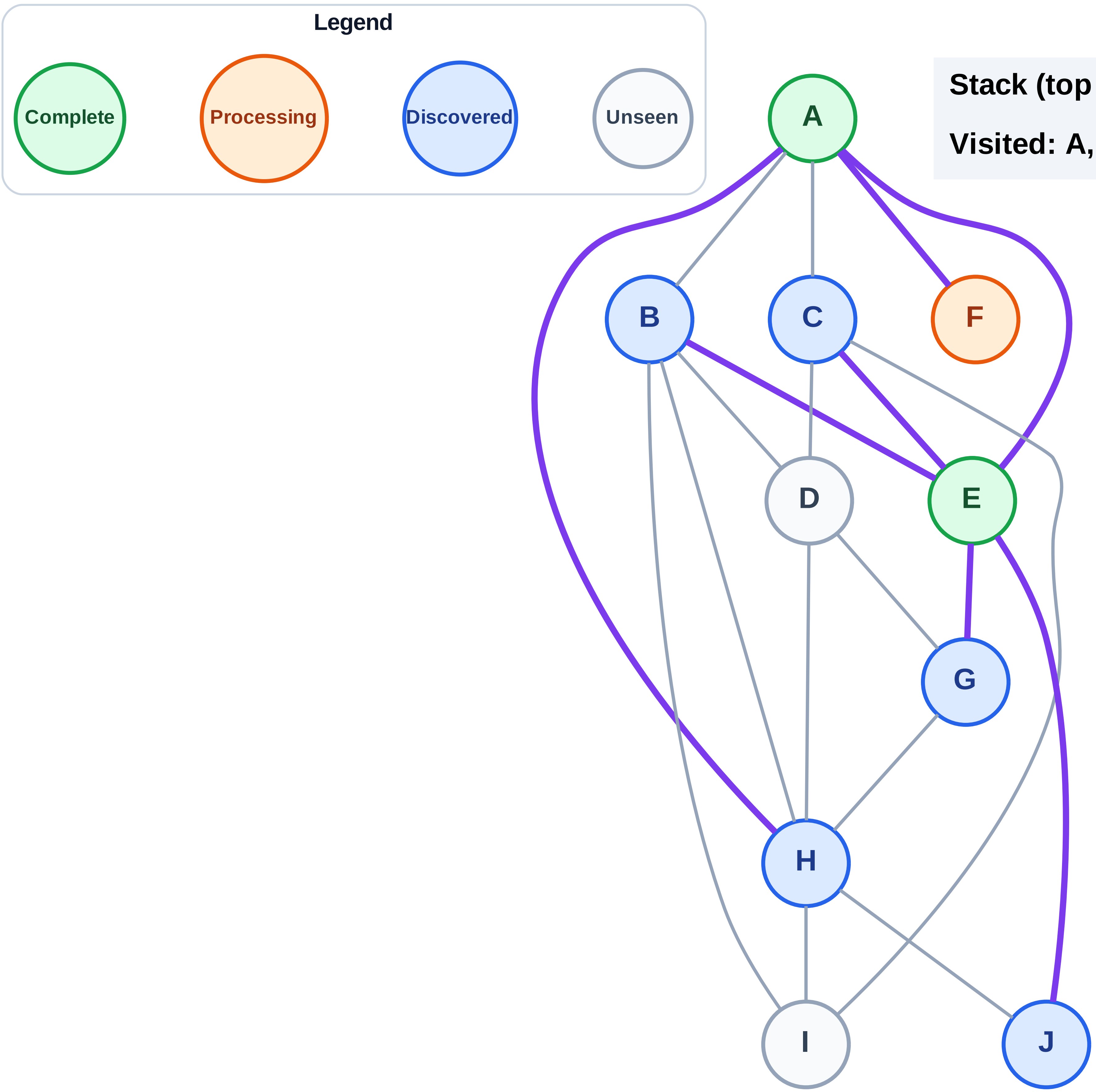


Stack (top at right): J | G | C | B | H | F

Visited: A, E

DFS Traversal

Step 6: Process node F



Stack (top at right): J | G | C | B | H

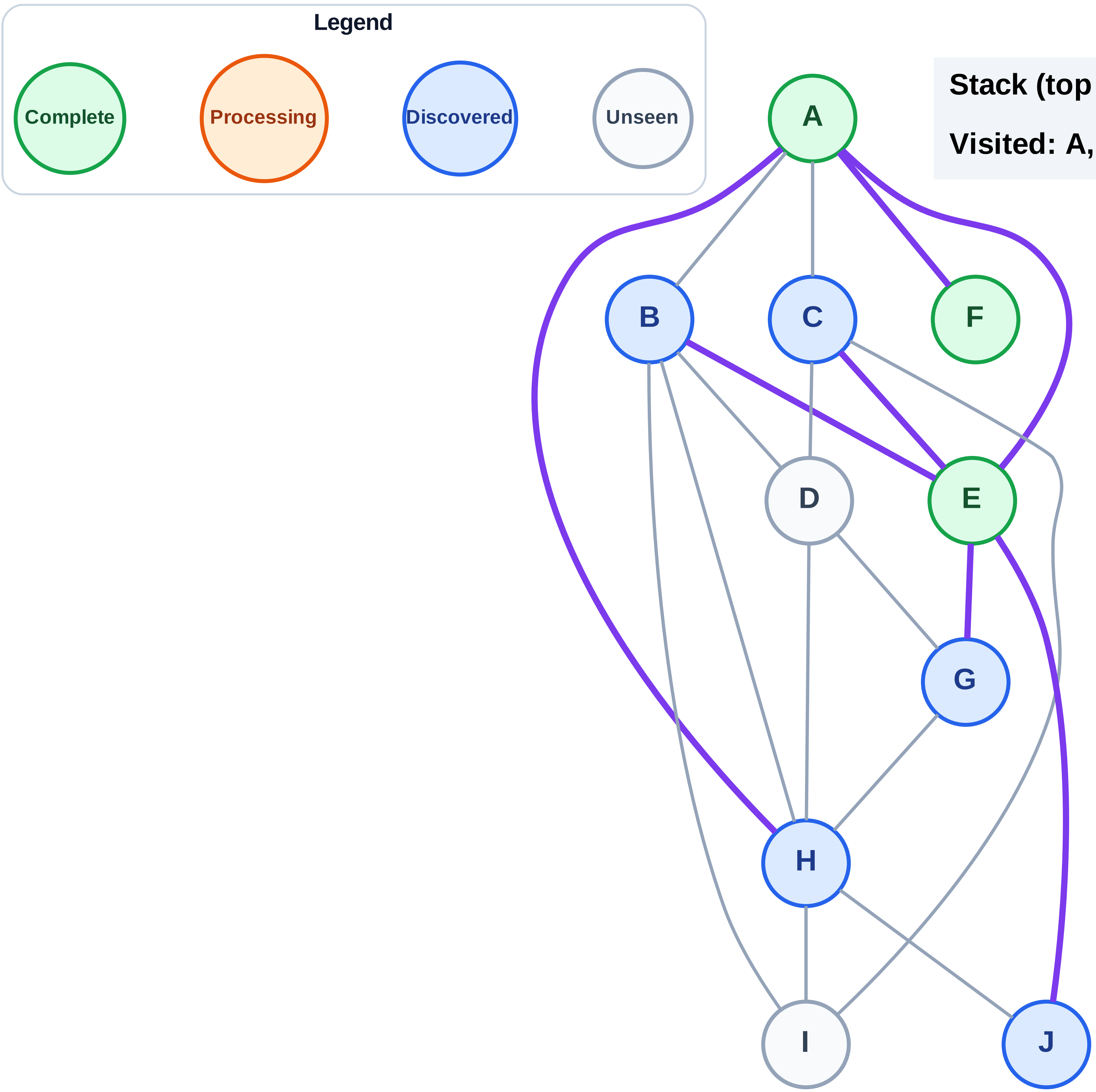
Visited: A, E

DFS Traversal

Step 7: No new neighbors from F

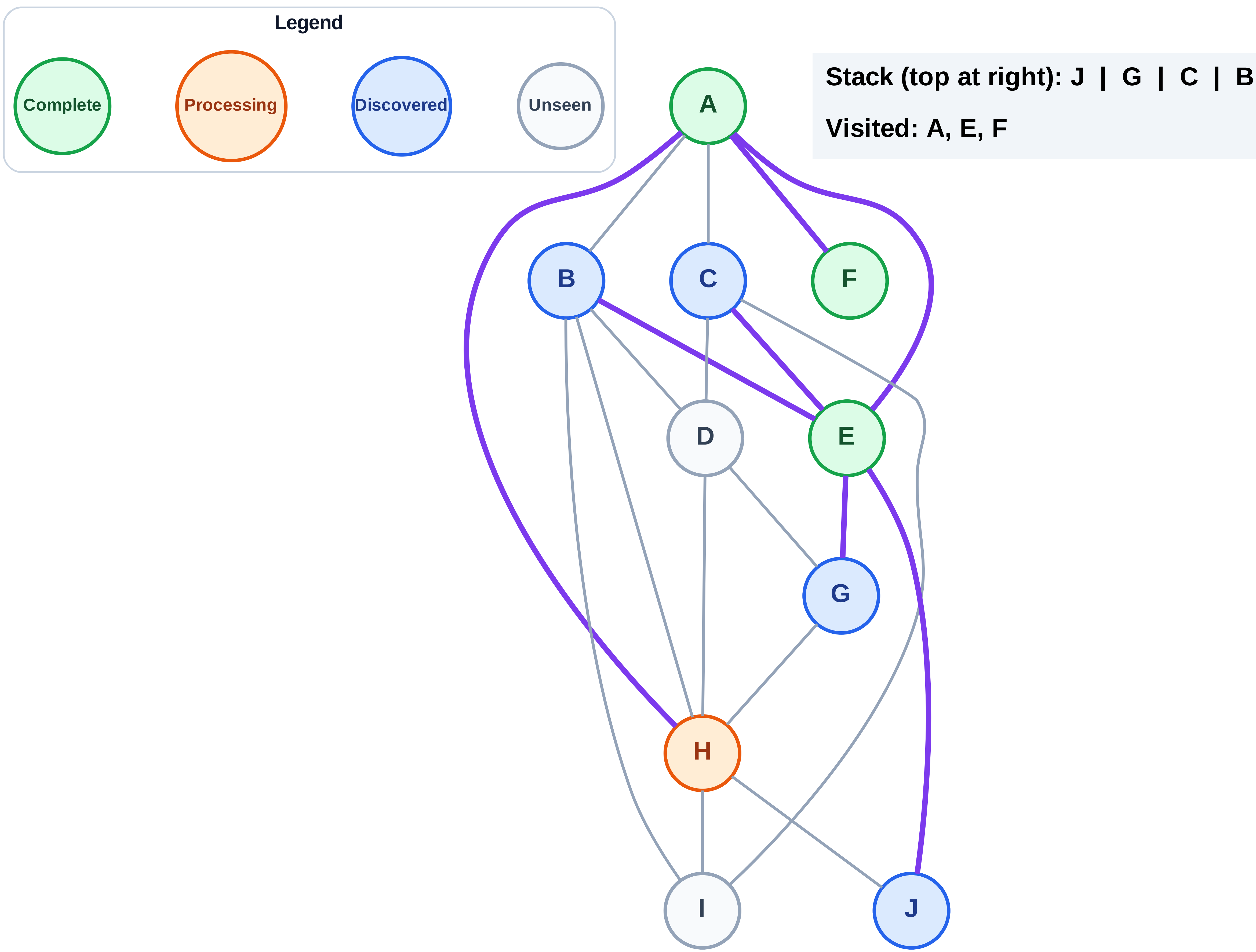
Stack (top at right): J | G | C | B | H

Visited: A, E, F



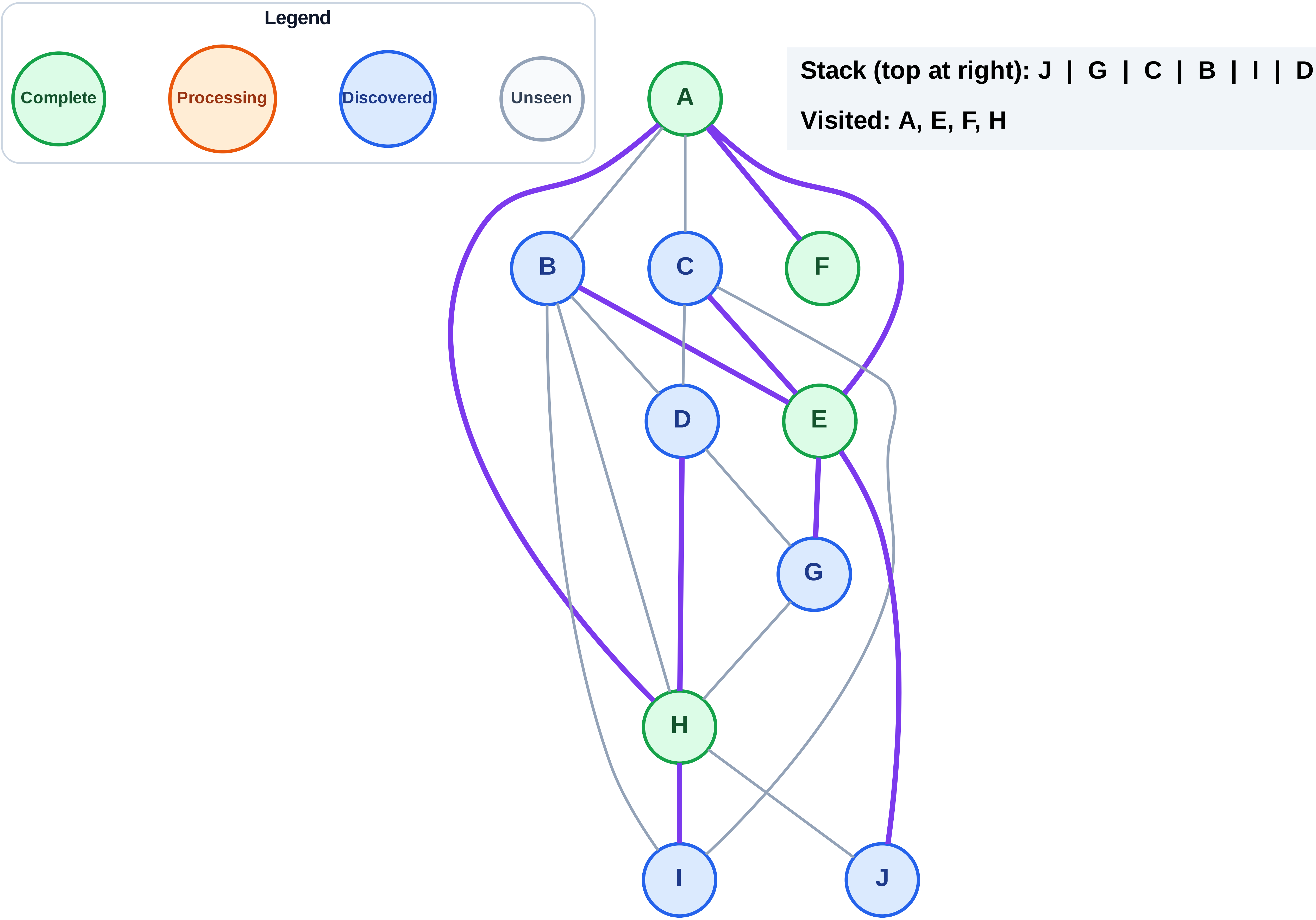
DFS Traversal

Step 8: Process node H



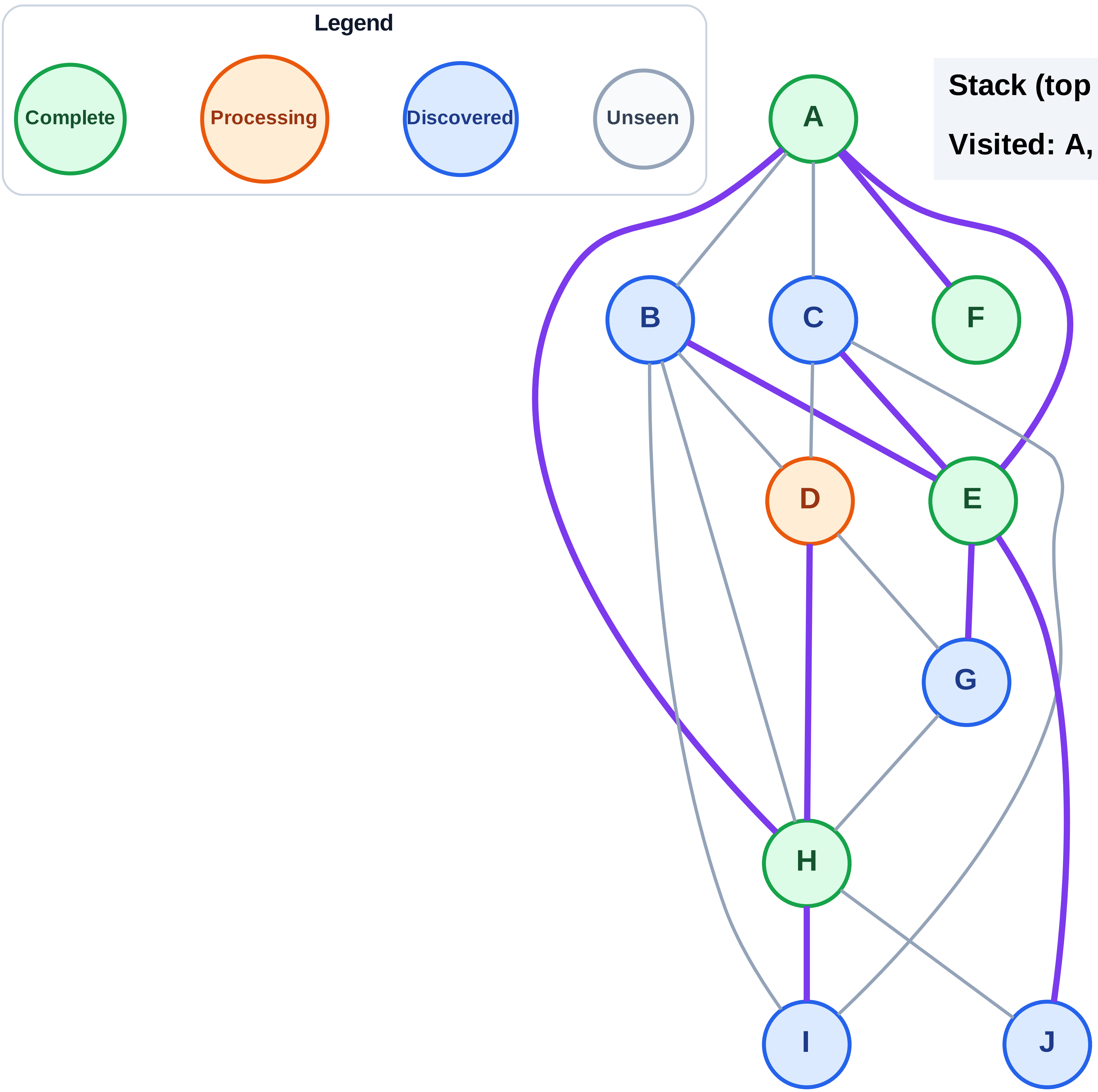
DFS Traversal

Step 9: Discover I, D from H



DFS Traversal

Step 10: Process node D

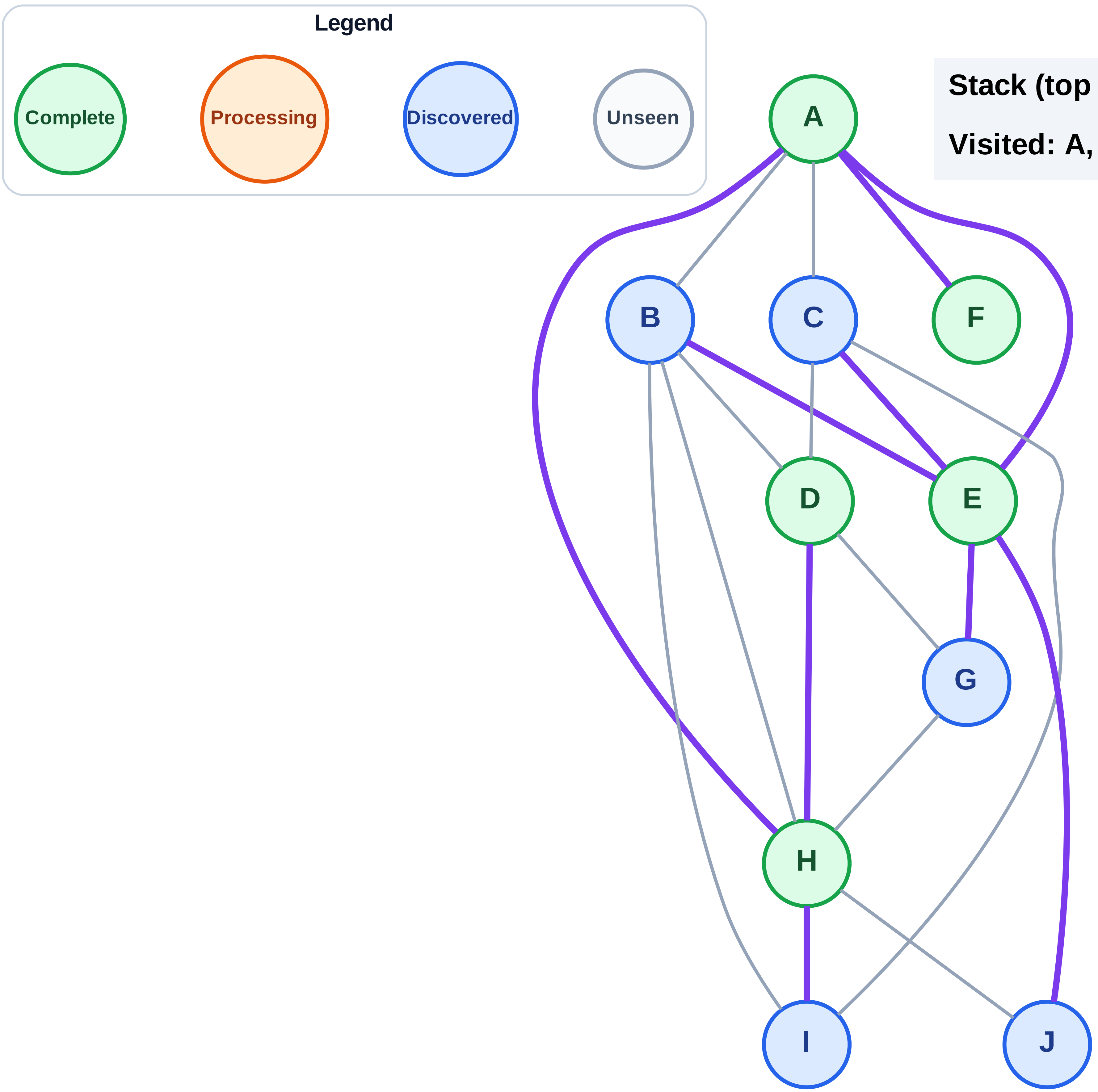


Stack (top at right): J | G | C | B | I

Visited: A, E, F, H

DFS Traversal

Step 11: No new neighbors from D

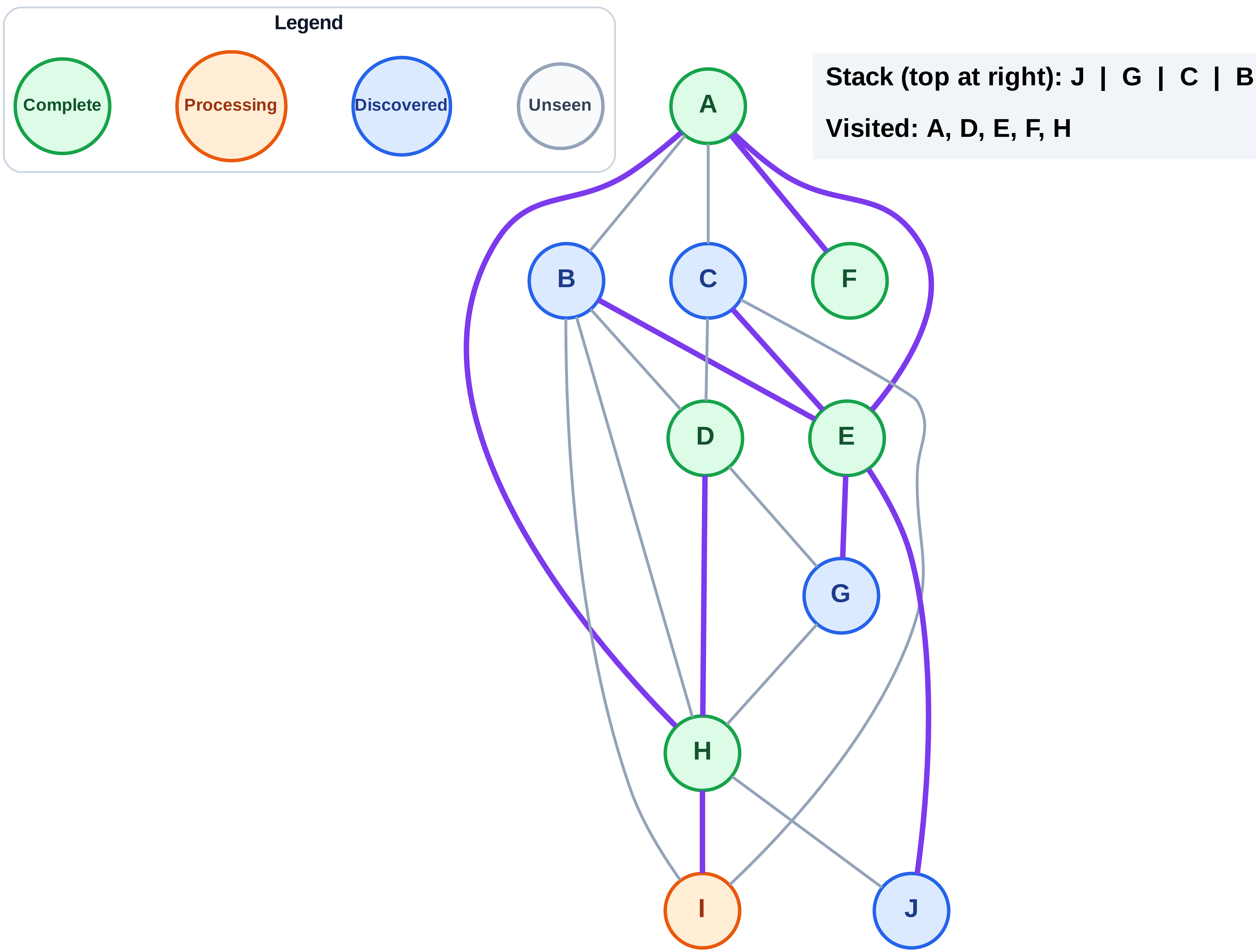


Stack (top at right): J | G | C | B | I

Visited: A, D, E, F, H

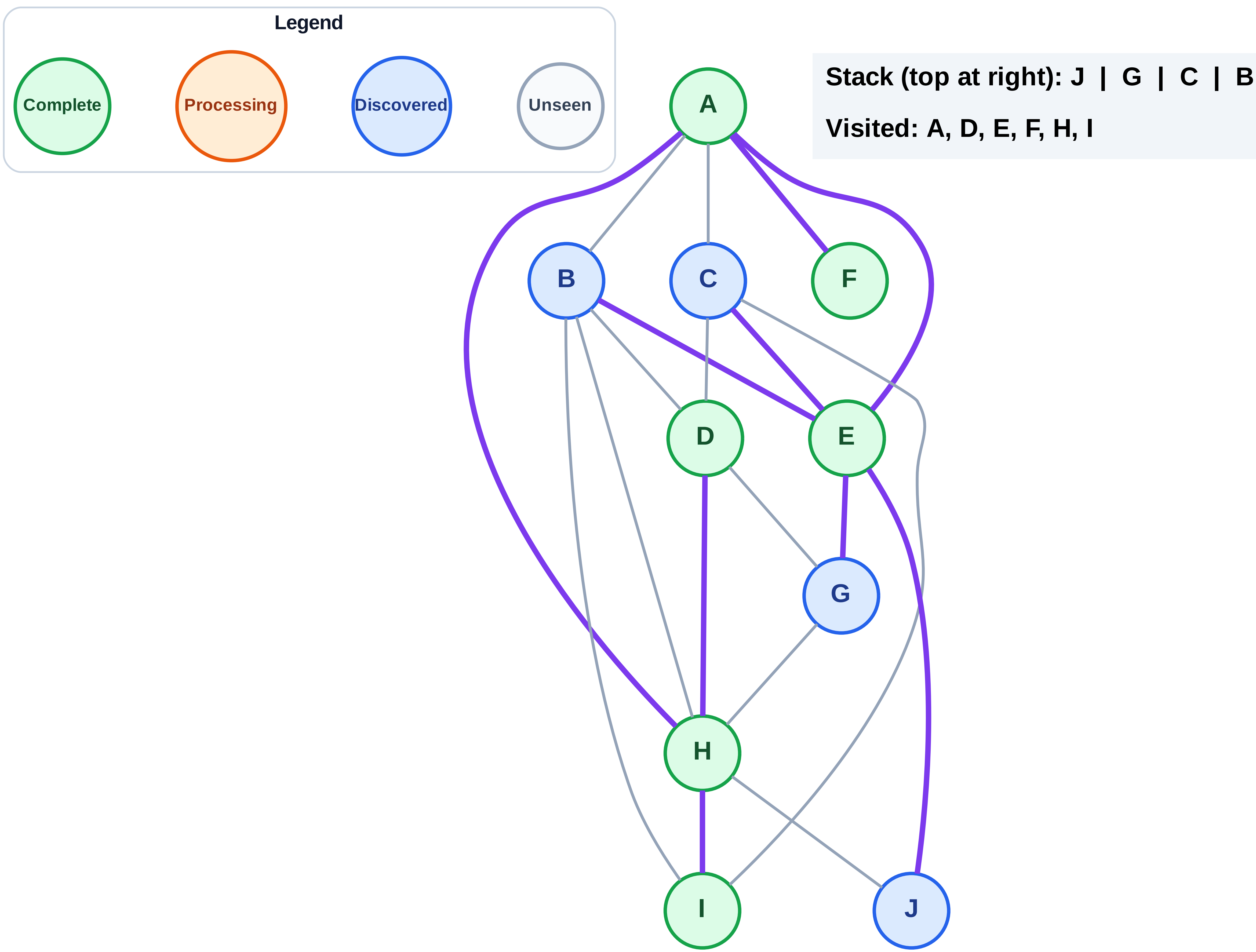
DFS Traversal

Step 12: Process node I



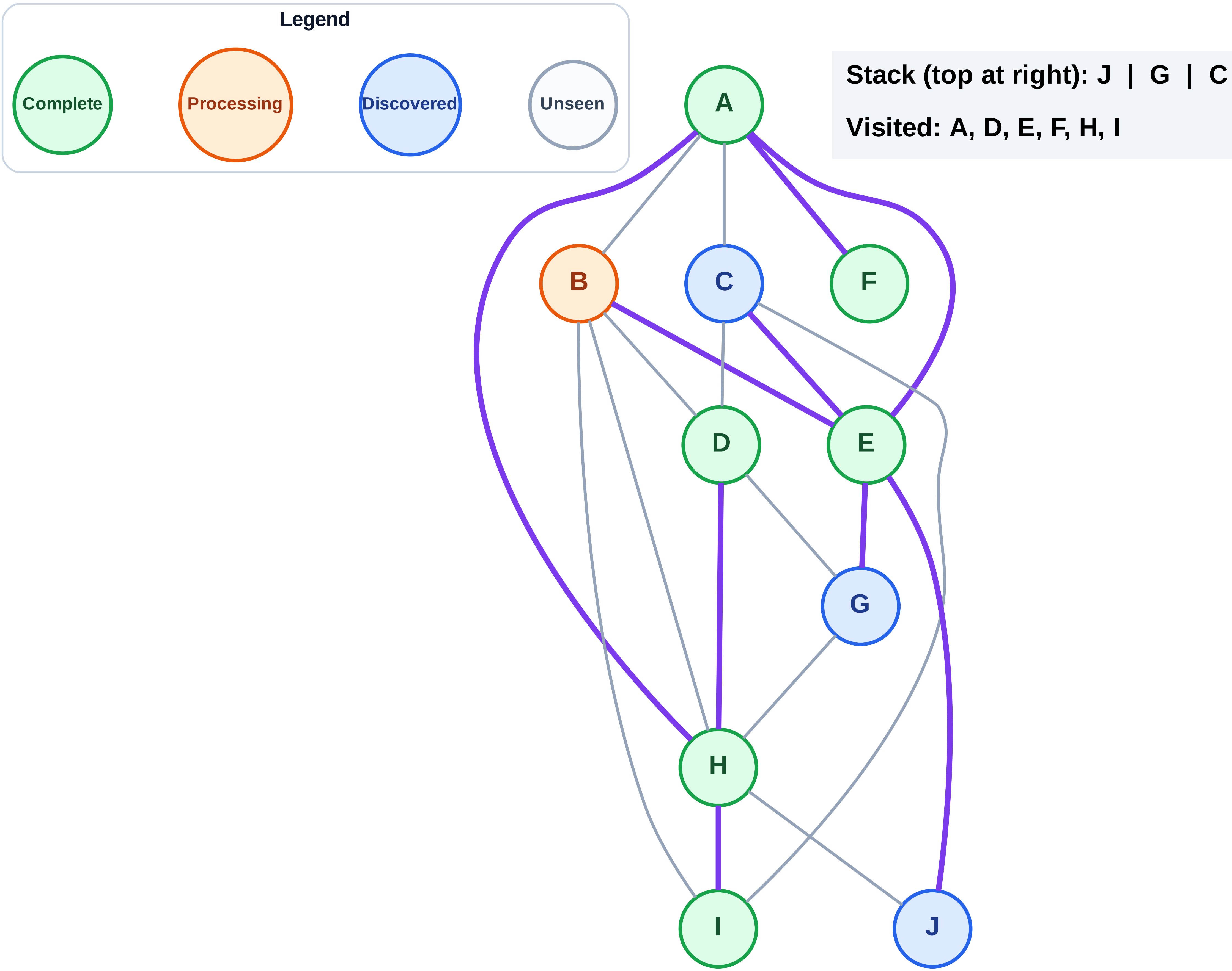
DFS Traversal

Step 13: No new neighbors from I



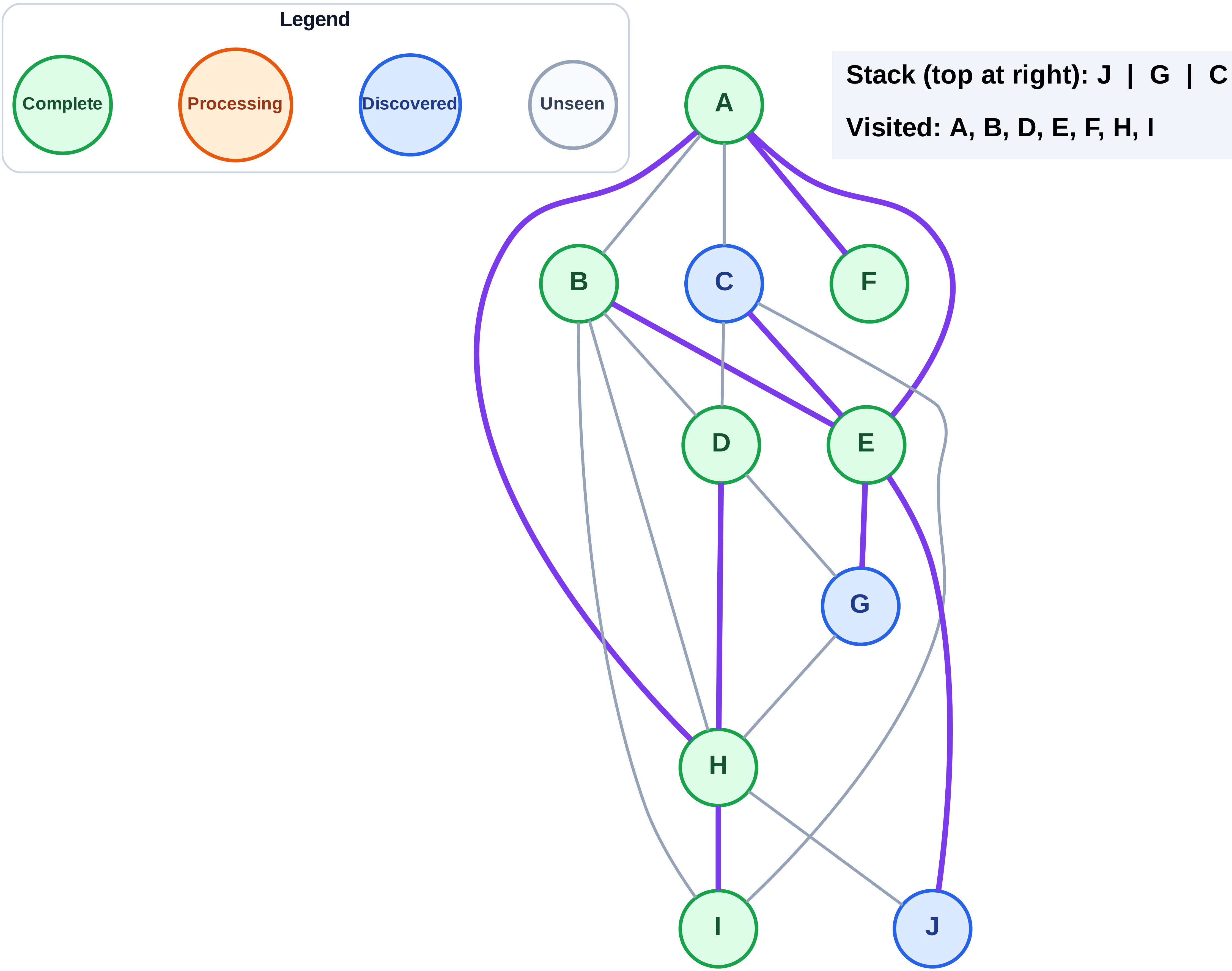
DFS Traversal

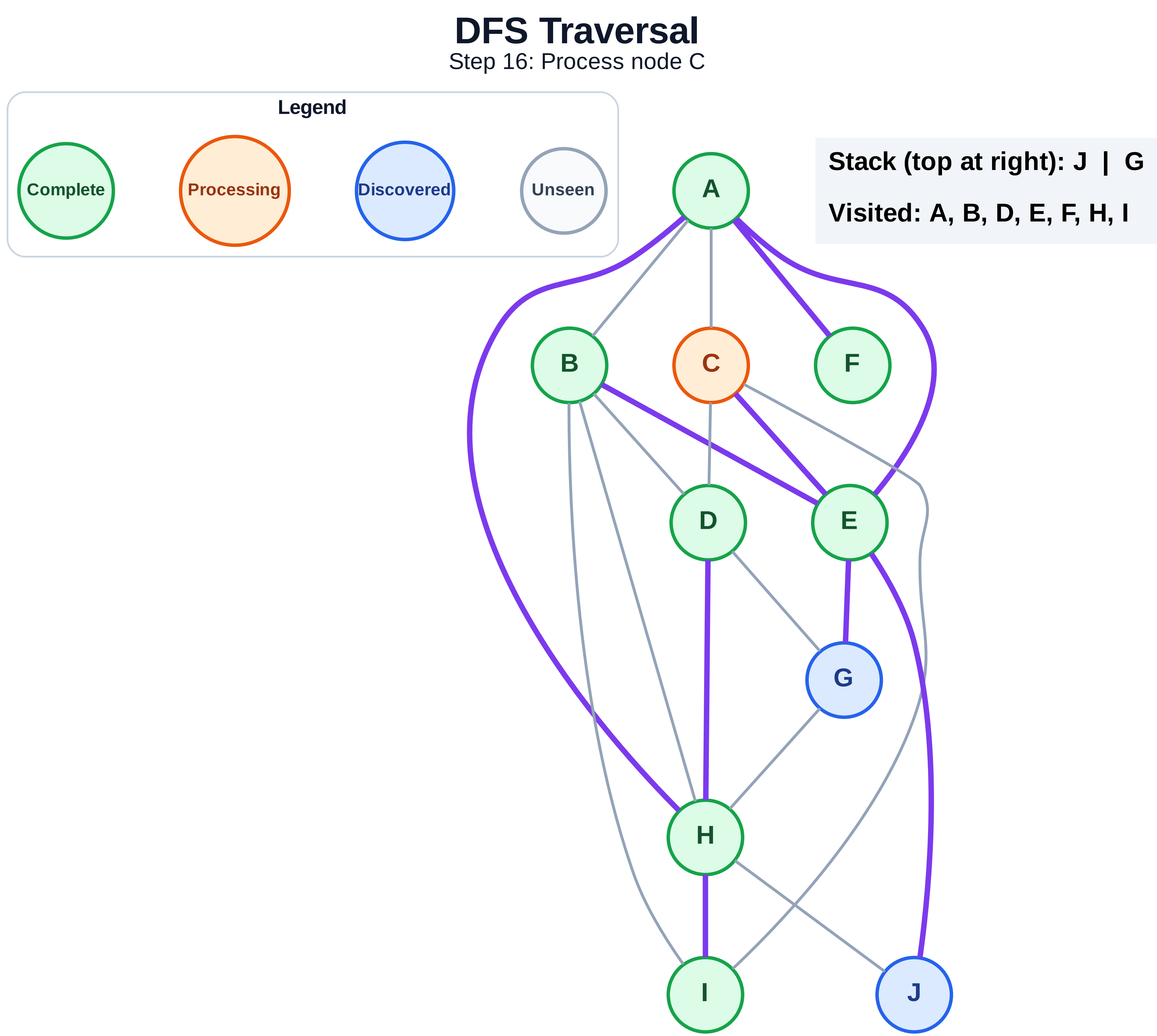
Step 14: Process node B



DFS Traversal

Step 15: No new neighbors from B

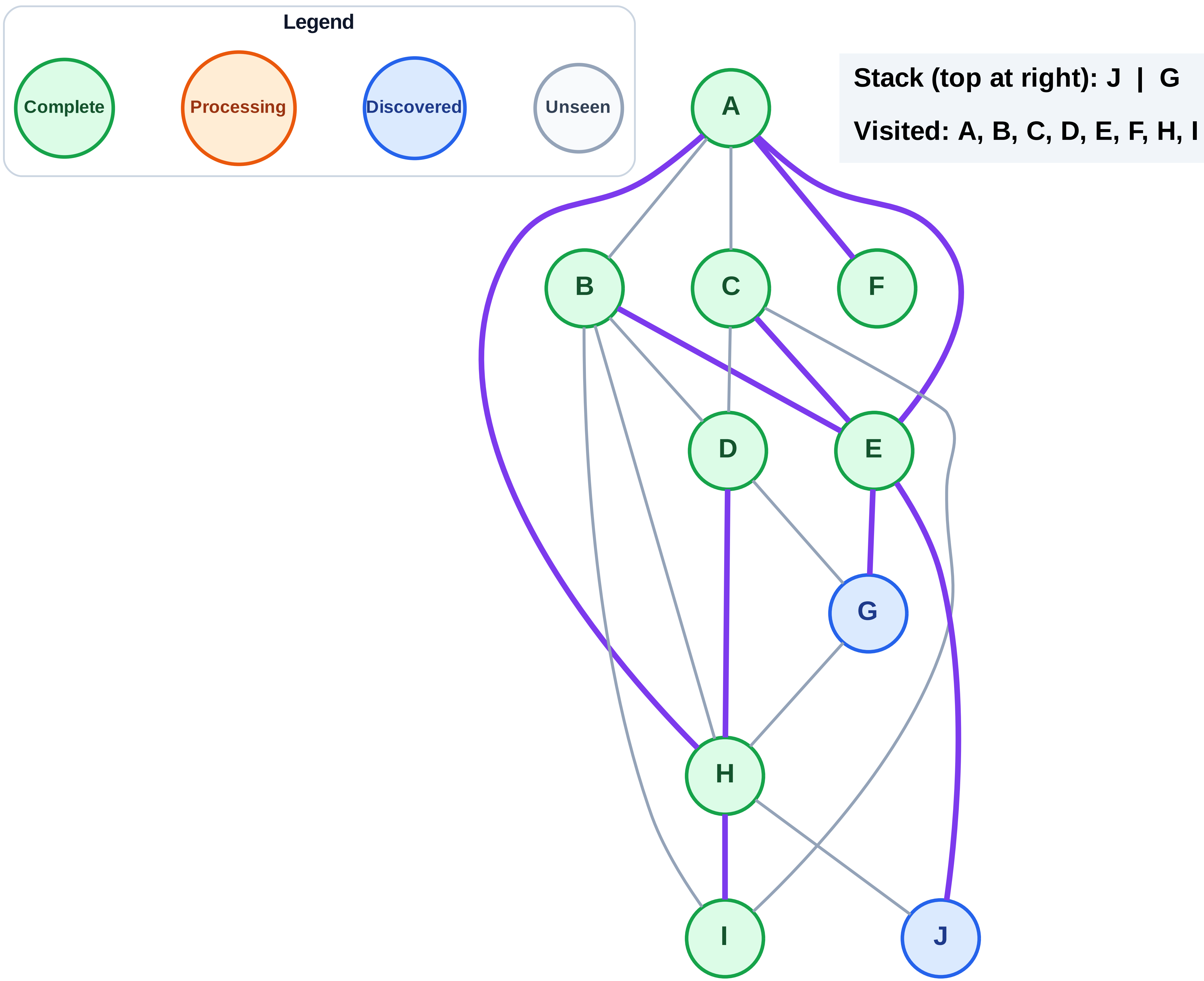




DFS Traversal
Step 16: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | G
Visited: A, B, D, E, F, H, I
A
B
C
F
D
E
G
H
I
J

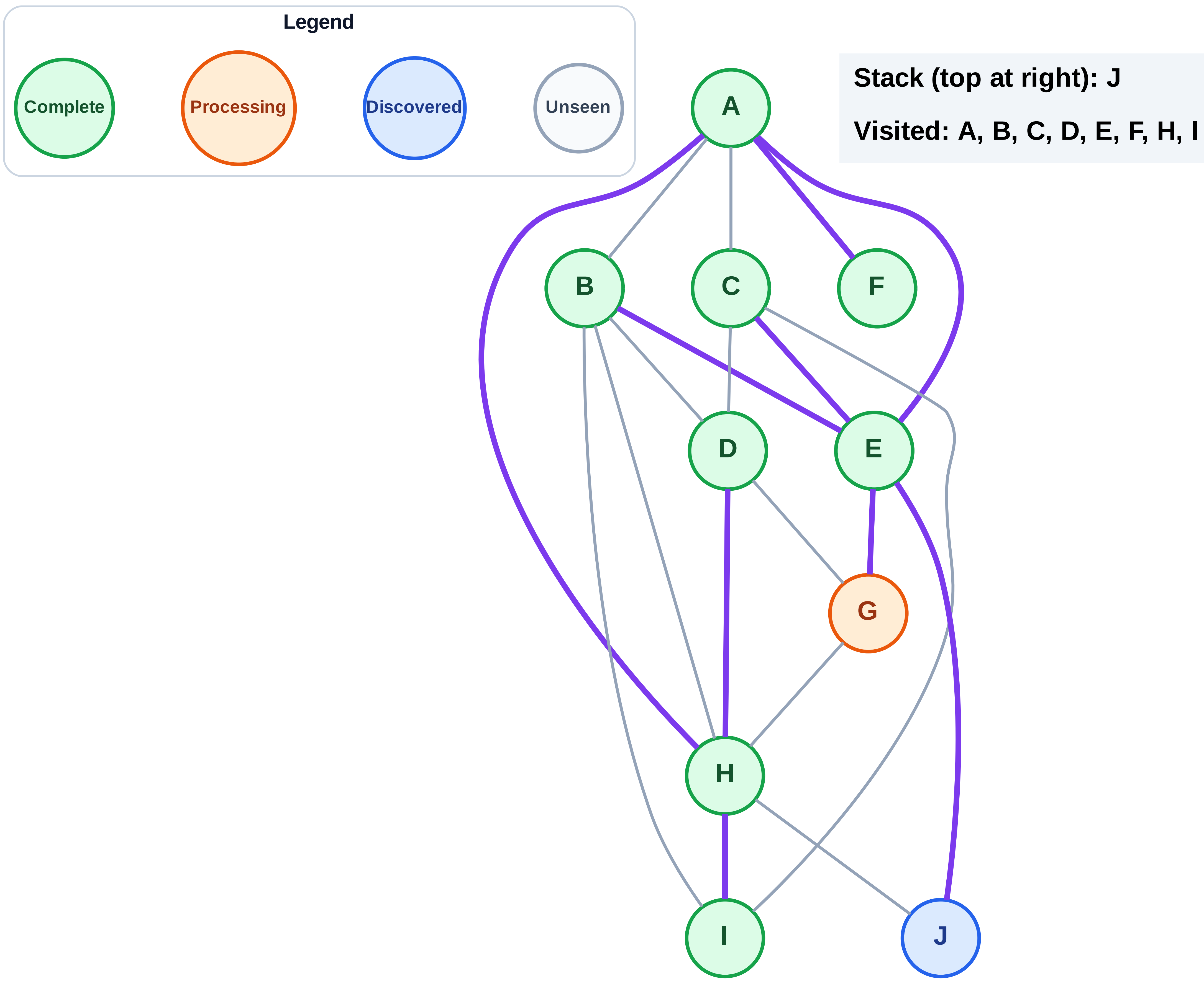
DFS Traversal

Step 17: No new neighbors from C



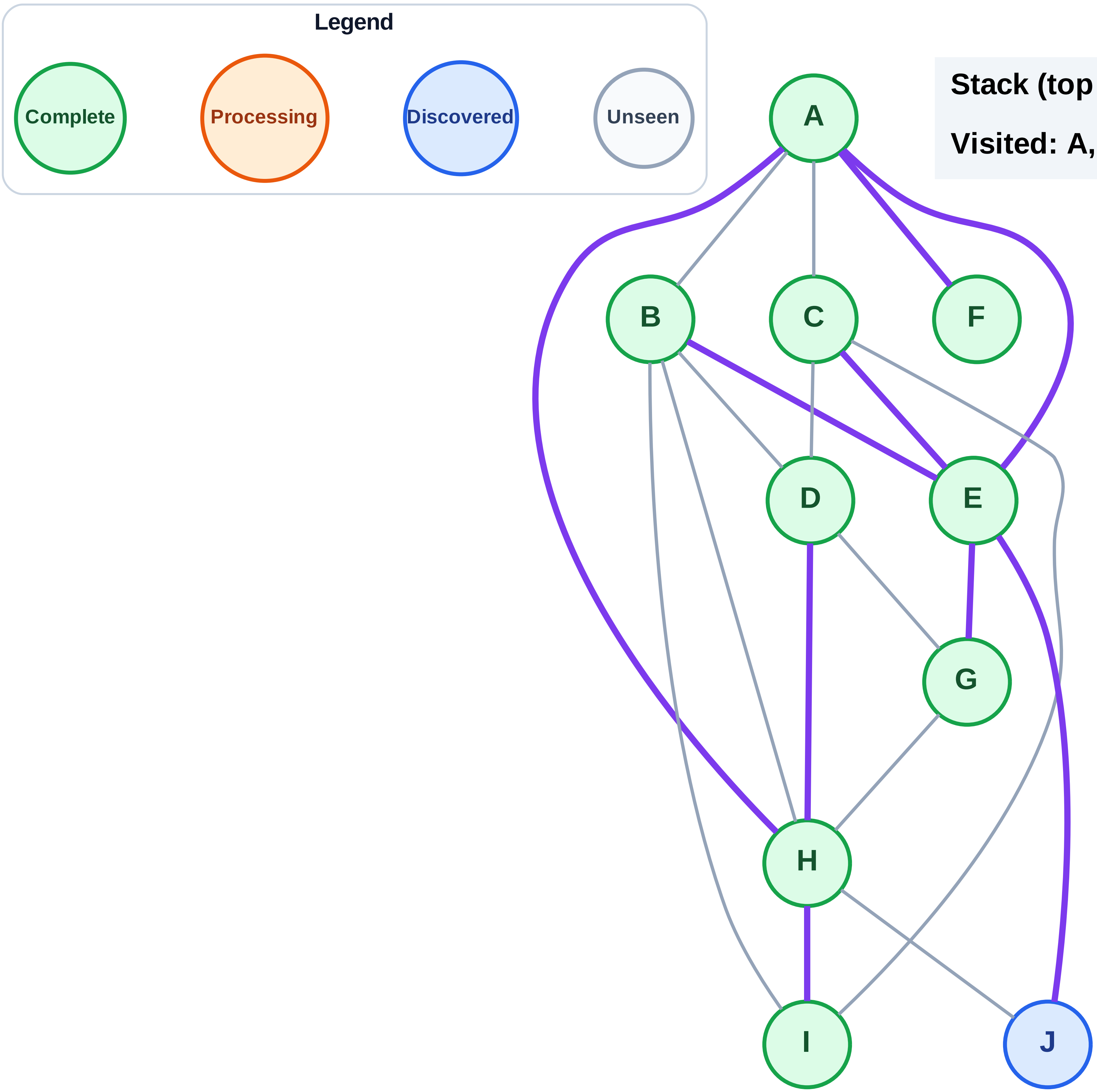
DFS Traversal

Step 18: Process node G



DFS Traversal

Step 19: No new neighbors from G

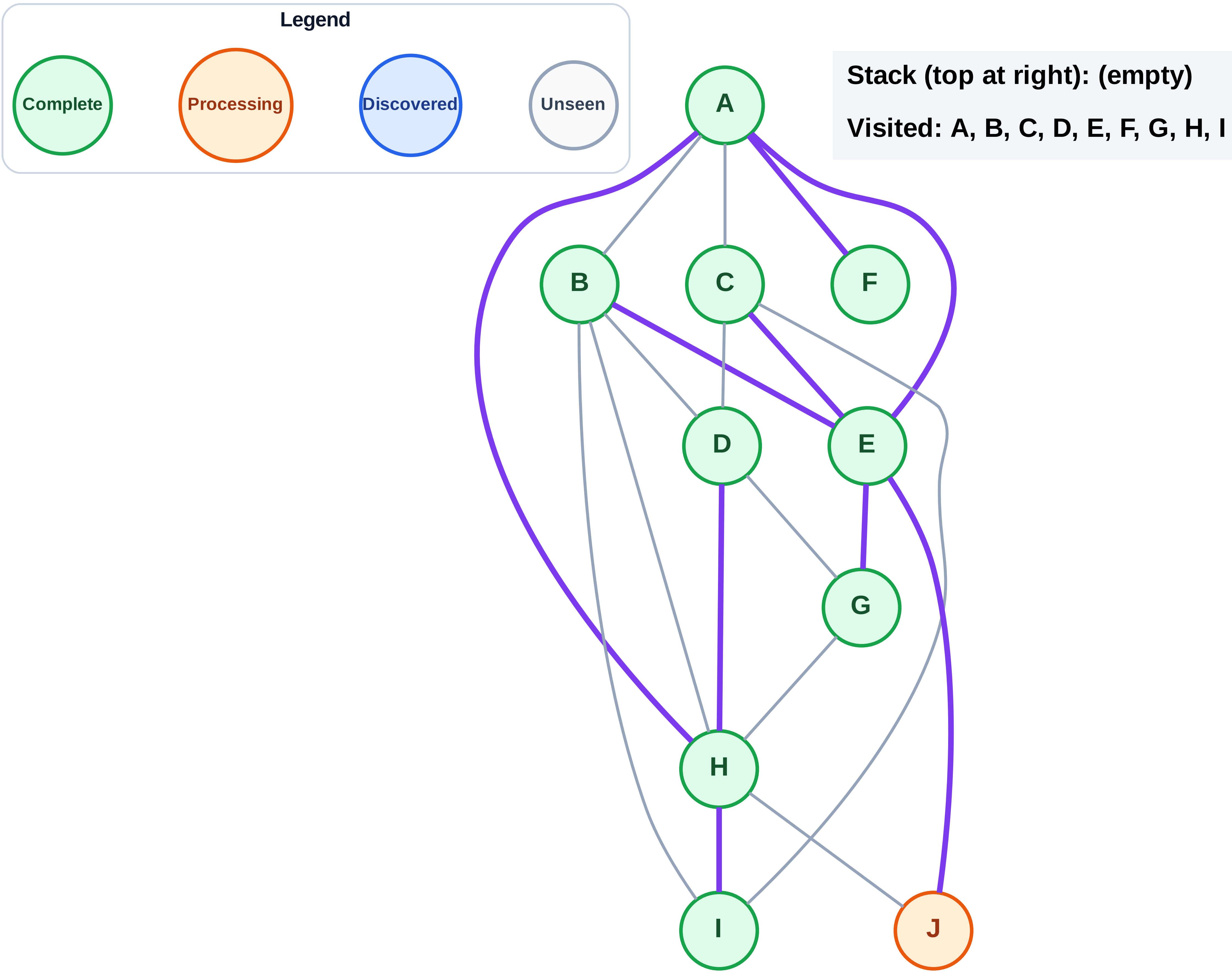


Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

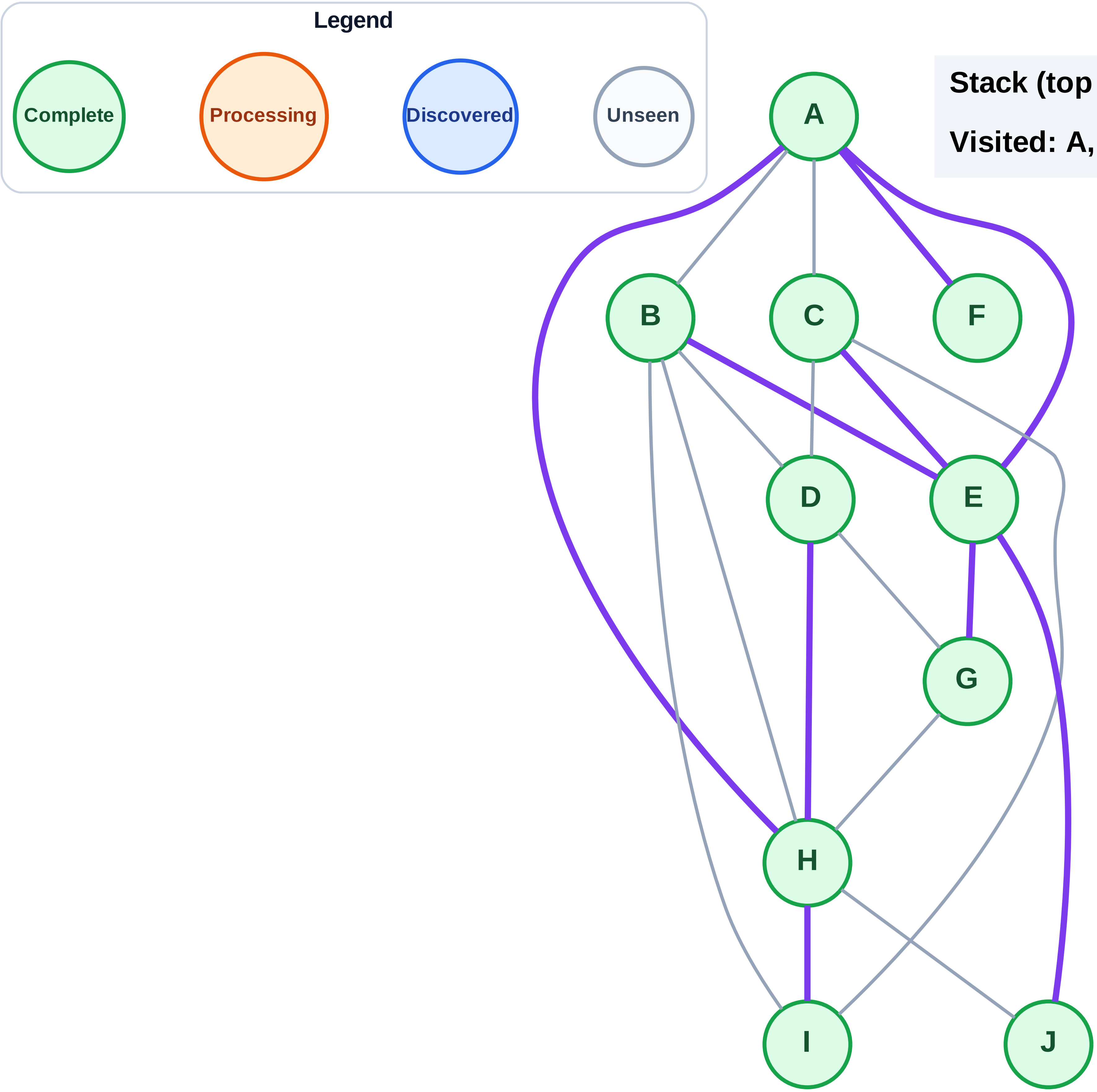
DFS Traversal

Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J